

IV. Graphs and Hypergraphs

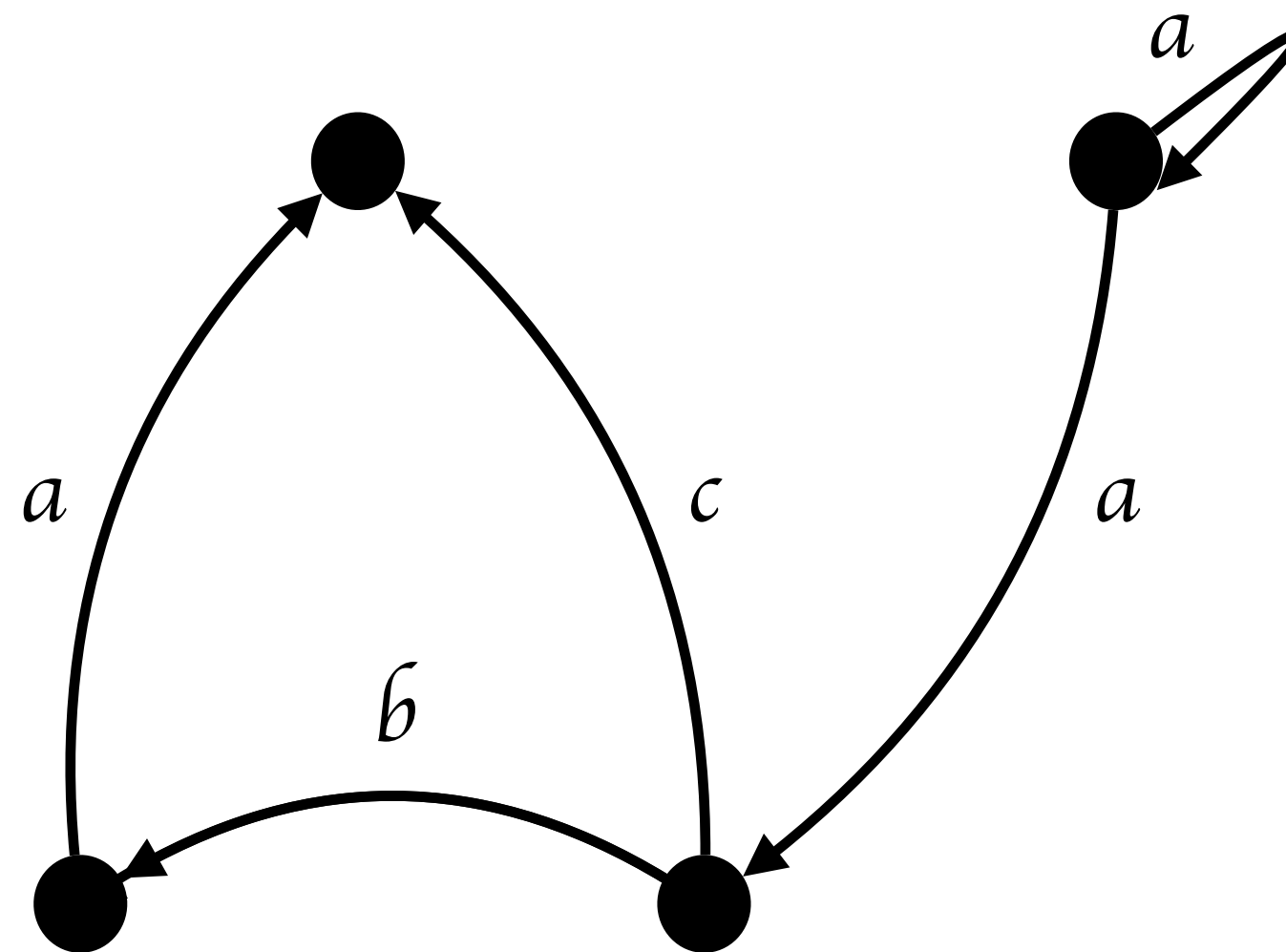
Radu Iosif (CNRS/VERIMAG)

Graphs

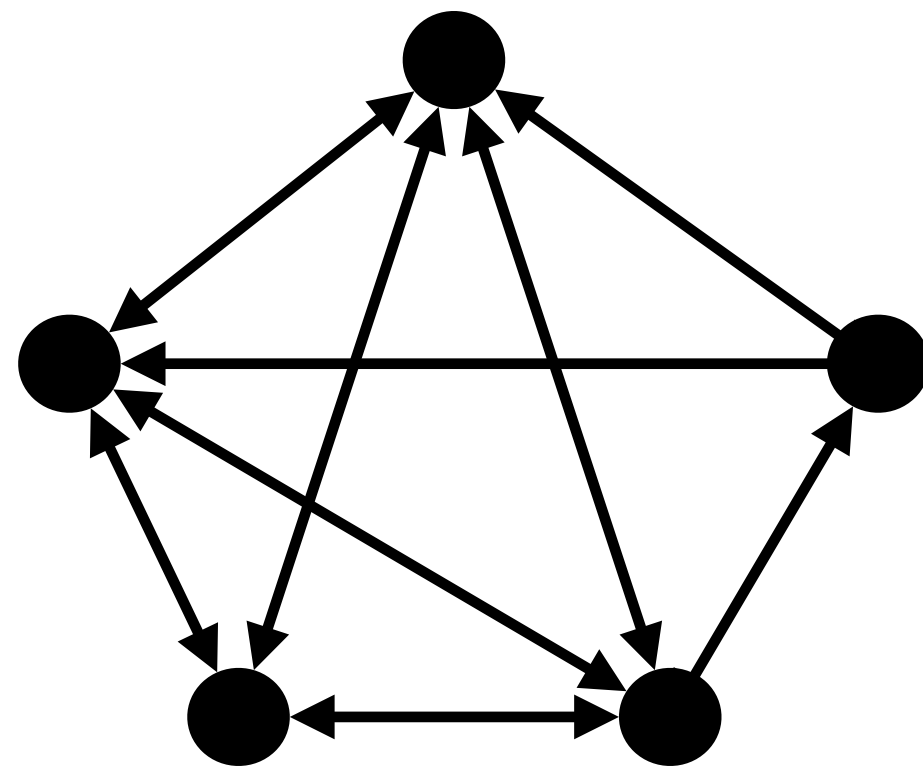
$\Sigma \stackrel{\text{def}}{=} \text{finite alphabet of binary edge labels}$

$V \stackrel{\text{def}}{=} \text{finite set of vertices}$

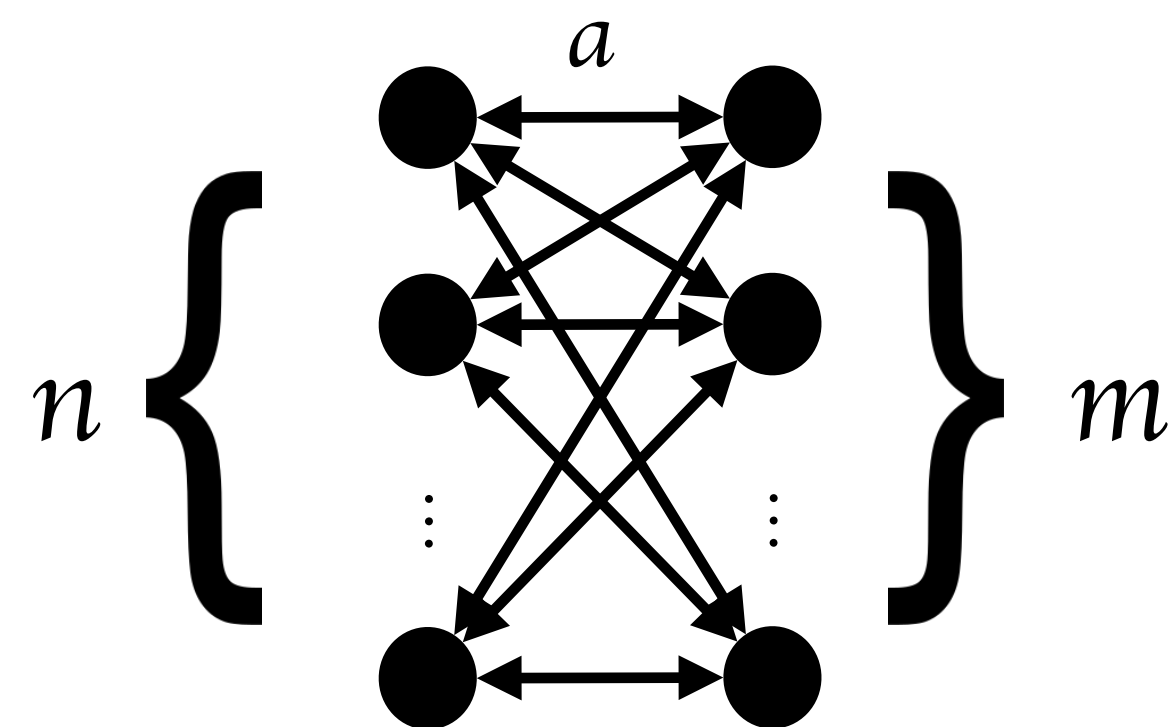
$E \subseteq V \times \Sigma \times V \stackrel{\text{def}}{=} \text{set of edges}$



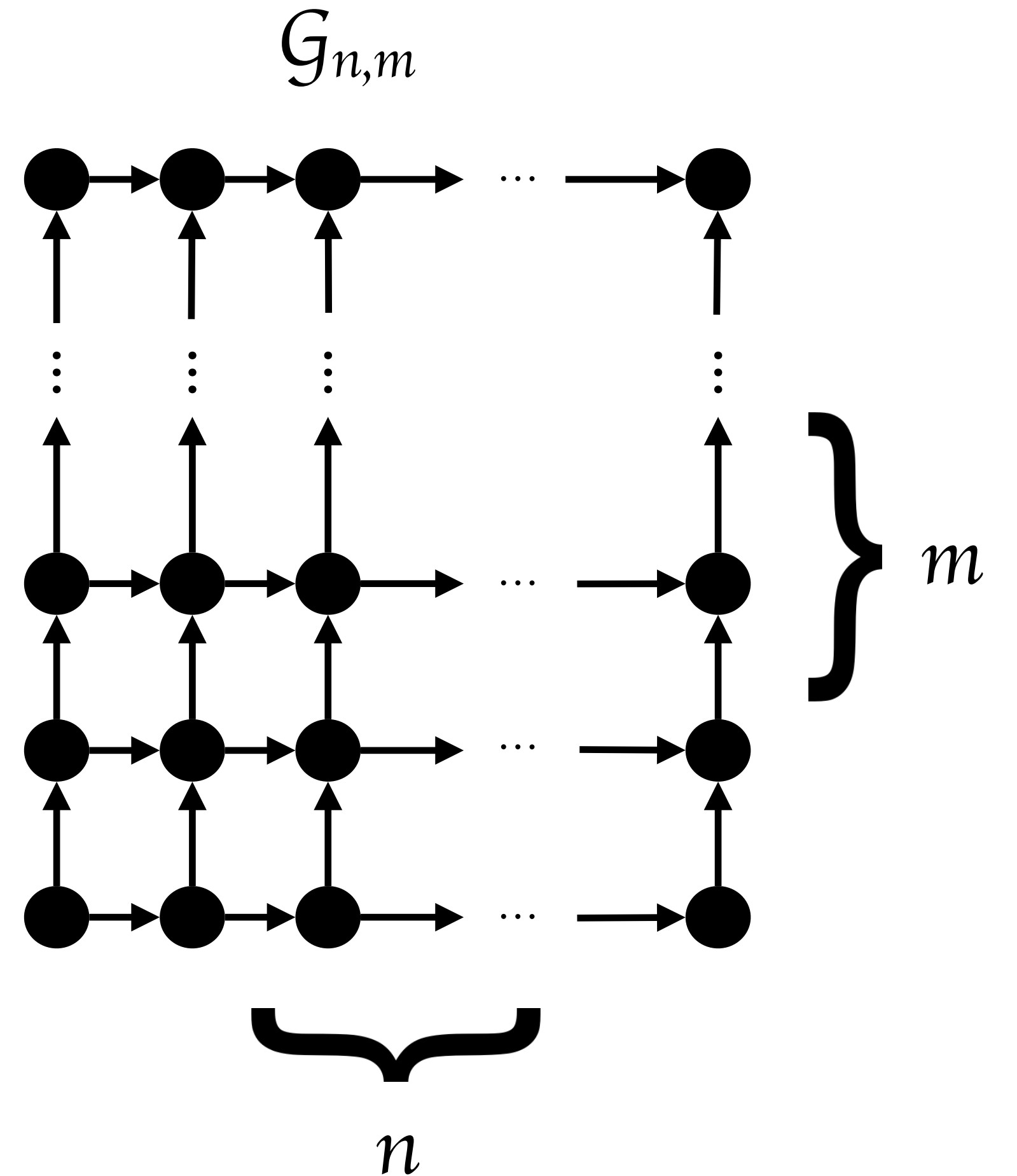
Cliques, Bipartite Graphs and Grids



K_5



$K_{n,m}$

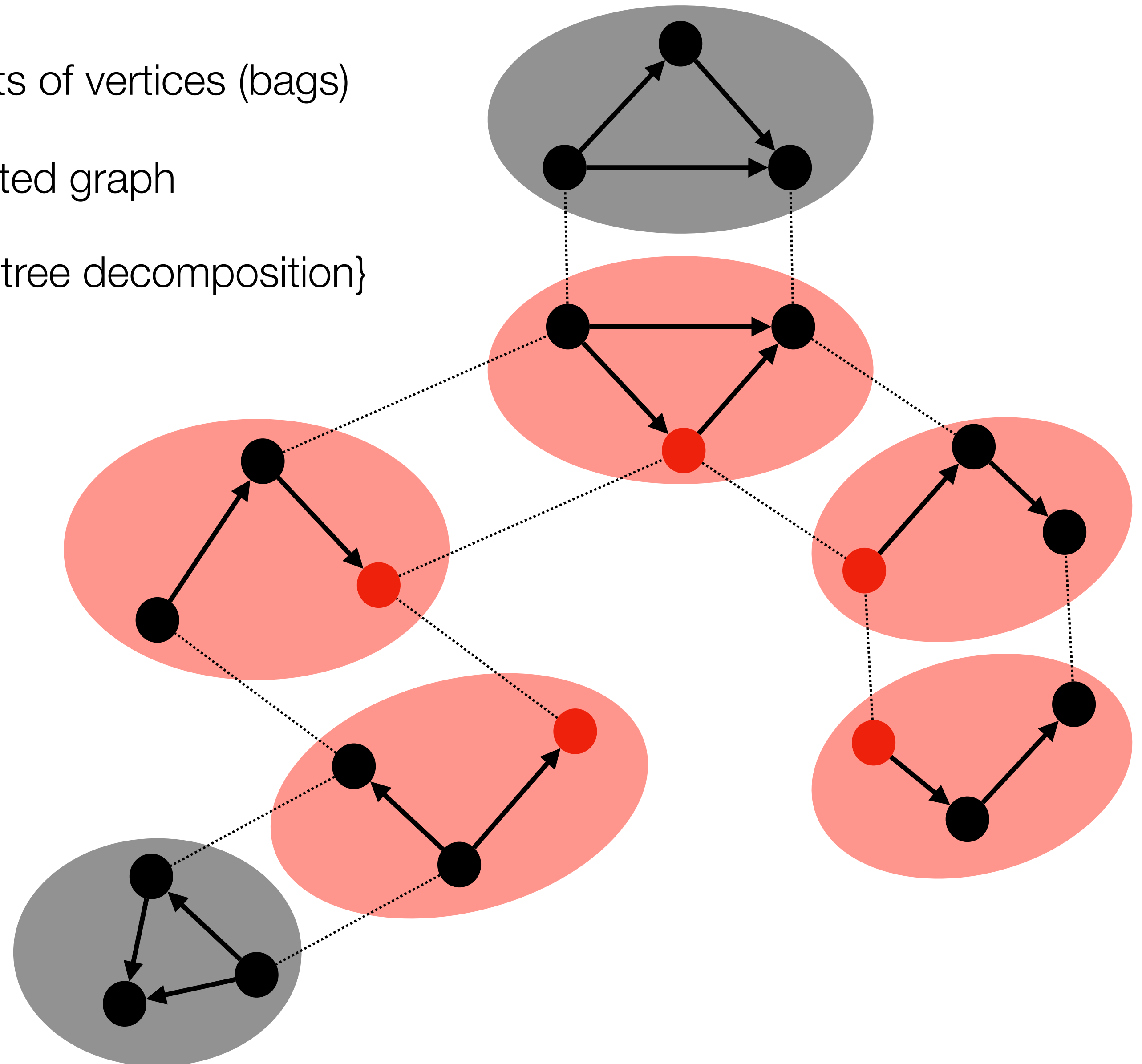
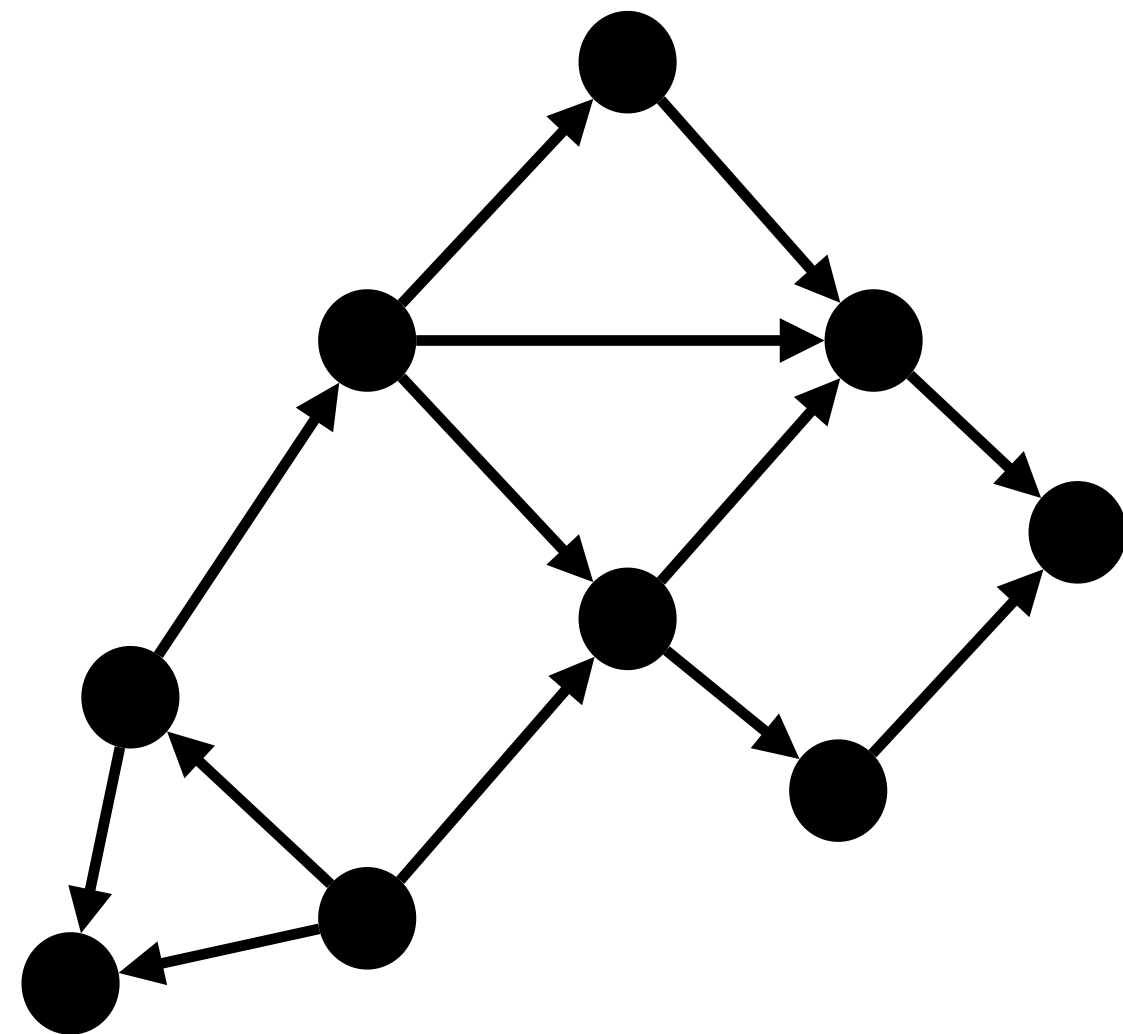


Tree-width

Tree decomposition $\stackrel{\text{def}}{=}$ unranked tree labeled with sets of vertices (bags)

1. each edge has both endpoints in a bag
2. the bags containing a given vertex form a connected graph

Tree-width $\stackrel{\text{def}}{=}$ $\min\{ \max\{ ||\text{bag}(n)|| \mid n \in V_T \} - 1 \mid T \text{ is a tree decomposition} \}$

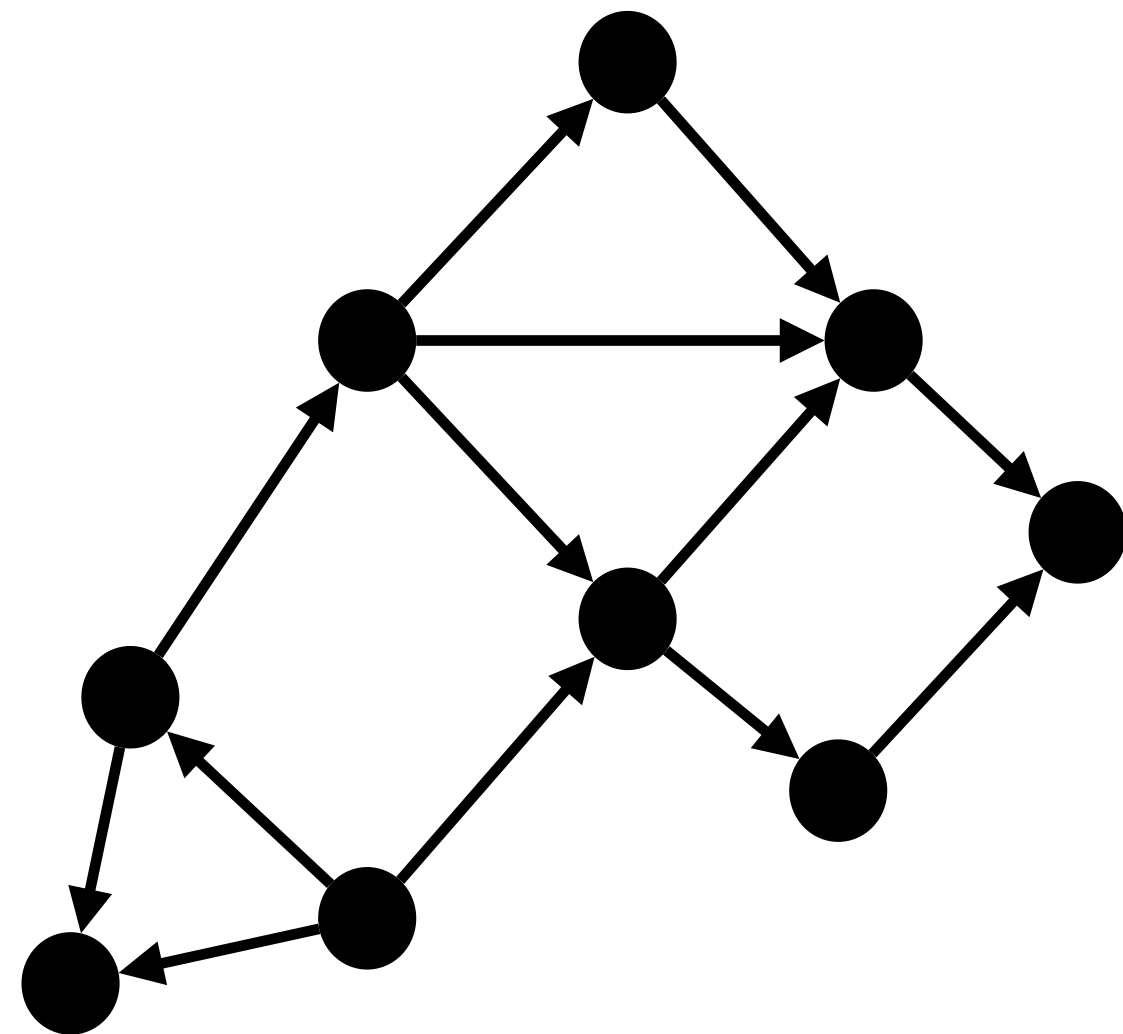


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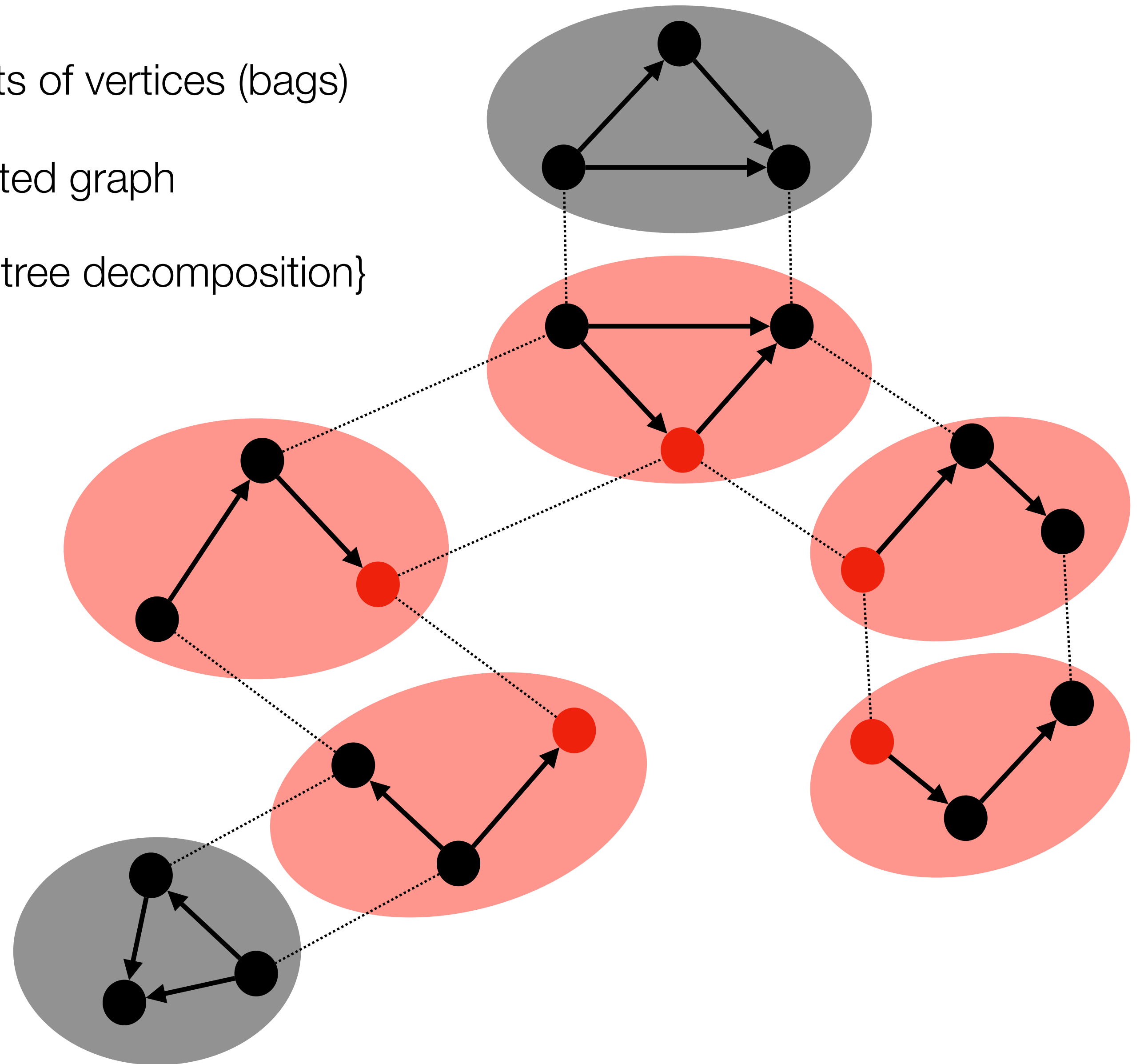
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$$tw(\mathcal{K}_n) = n-1$$

$$tw(G_{n,m}) = \min(n, m)$$



Logics for Graphs

$\Sigma \stackrel{\text{def}}{=} \text{finite alphabet of binary edge labels}$

$a(x,y)$	$\exists x$	$\exists \mathcal{X}$
there exists an edge from x to y with label $a \in \Sigma$	there exists a vertex x	there exists a set of vertices $\mathcal{X}$


the **adjacency** model (CMSO₁)

CMSO = MSO with modulo cardinality constraints

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 the **adjacency** model (CMSO₁)

$edg_a(x,y,z)$	$\exists x$	$\exists \mathcal{X}$
x is an edge from y to z with label $a \in \Sigma$	there exists a vertex/edge x	there exists a set of vertices and edges $\mathcal{X}$


 the **incidence** model (CMSO₂)

CMSO = MSO with modulo cardinality constraints

Properties of Graphs Definable in MSO

$$\exists X_1 \dots \exists X_k . \underbrace{\bigwedge_{1 \leq i < j \leq k} X_i \cap X_j = \emptyset}_{\text{no two vertices have the same color}} \wedge \forall x . \bigvee_{1 \leq i \leq k} x \in X_i \wedge \forall y . \bigwedge_{a \in \Sigma} a(x, y) \Rightarrow y \notin X_i \quad \text{MSO}_1$$

k -Colorability: each vertex can be colored with one of k colors, no adjacent vertices having the same color

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$\varphi(x, y, X)$ is the conjunction of :

1. x and y are vertices and $x \neq y$
2. X is a set of edges
3. x (y) is the head (tail) of exactly one edge in X
4. any other vertex incident to some edge in X is the head and the tail of exactly one edge in X

MSO₂

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$$\psi(x, y, X) \stackrel{\text{def}}{=} \underbrace{\varphi(x, y, X)}_{\text{MSO}_2} \wedge \forall Y \subseteq X . \neg \varphi(x, y, Y)$$

there is an **elementary** path from x to y traversing the edges in X

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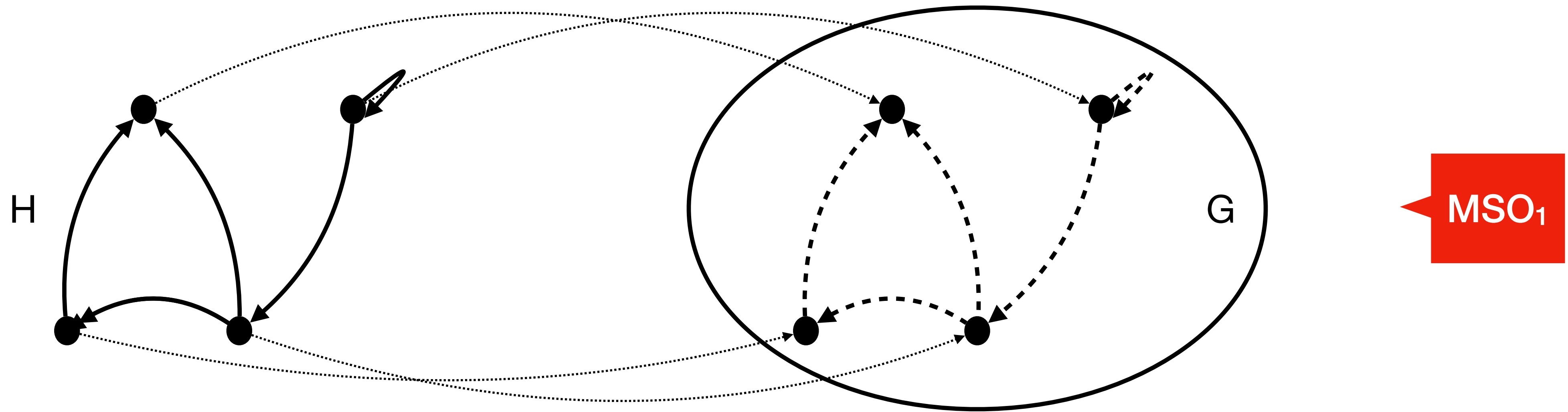
$$\exists x \exists y . \underbrace{\psi(x, y, X) \wedge \forall x . x \text{ is incident to some edge in } X \wedge \exists z . \text{edg}_a(z, y, x)}_{\text{Hamiltonicity}}$$

Hamiltonicity: there exists a circuit the traverses each vertex once

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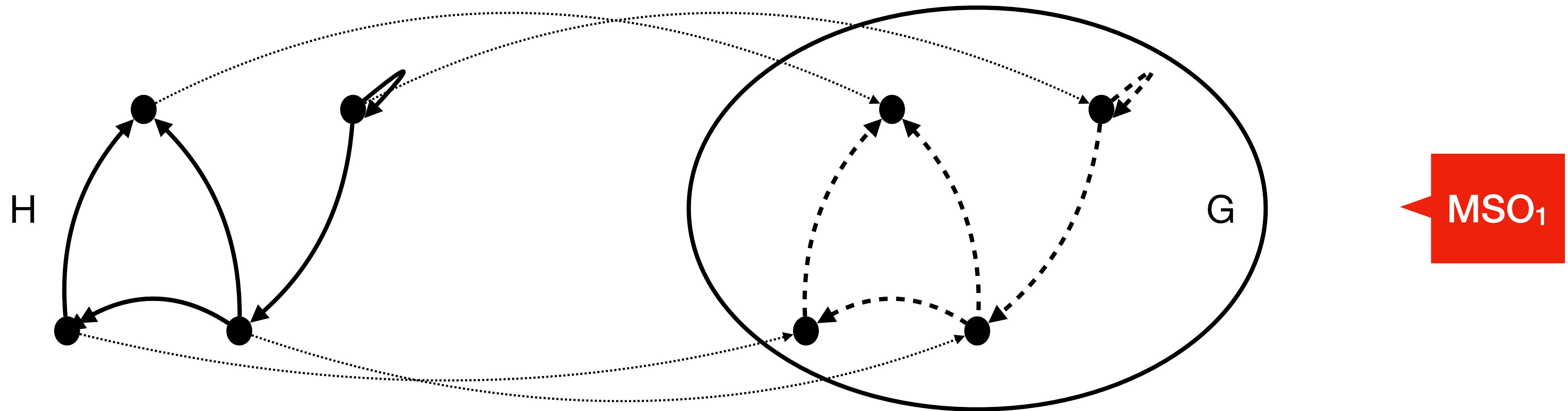
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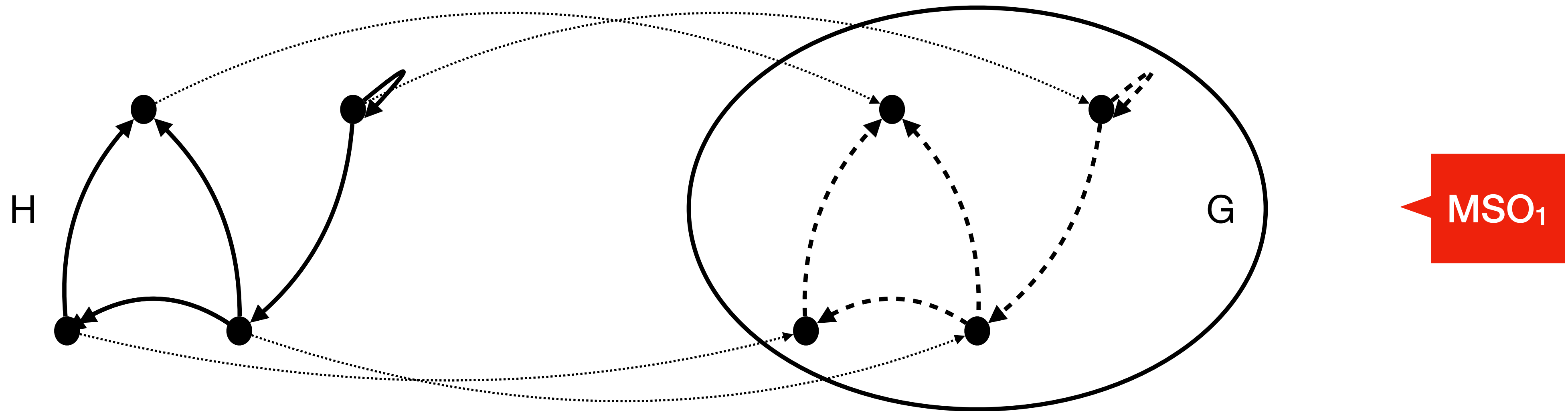


H is a **minor** of G $\stackrel{\text{def}}{=}$ H is obtained from G by contracting edges and deleting edges and vertices

Theorem [Robertson&Seymour] The minor relation is a **well-quasi-order**. As a consequence, the set $\{G \mid tw(G) \leq k\}$, for any $k \geq 1$, is defined by a finite set of excluded minors.

Example: $\mathcal{K}_4$ is the only excluded minor of the set $\{G \mid tw(G) \leq 2\}$

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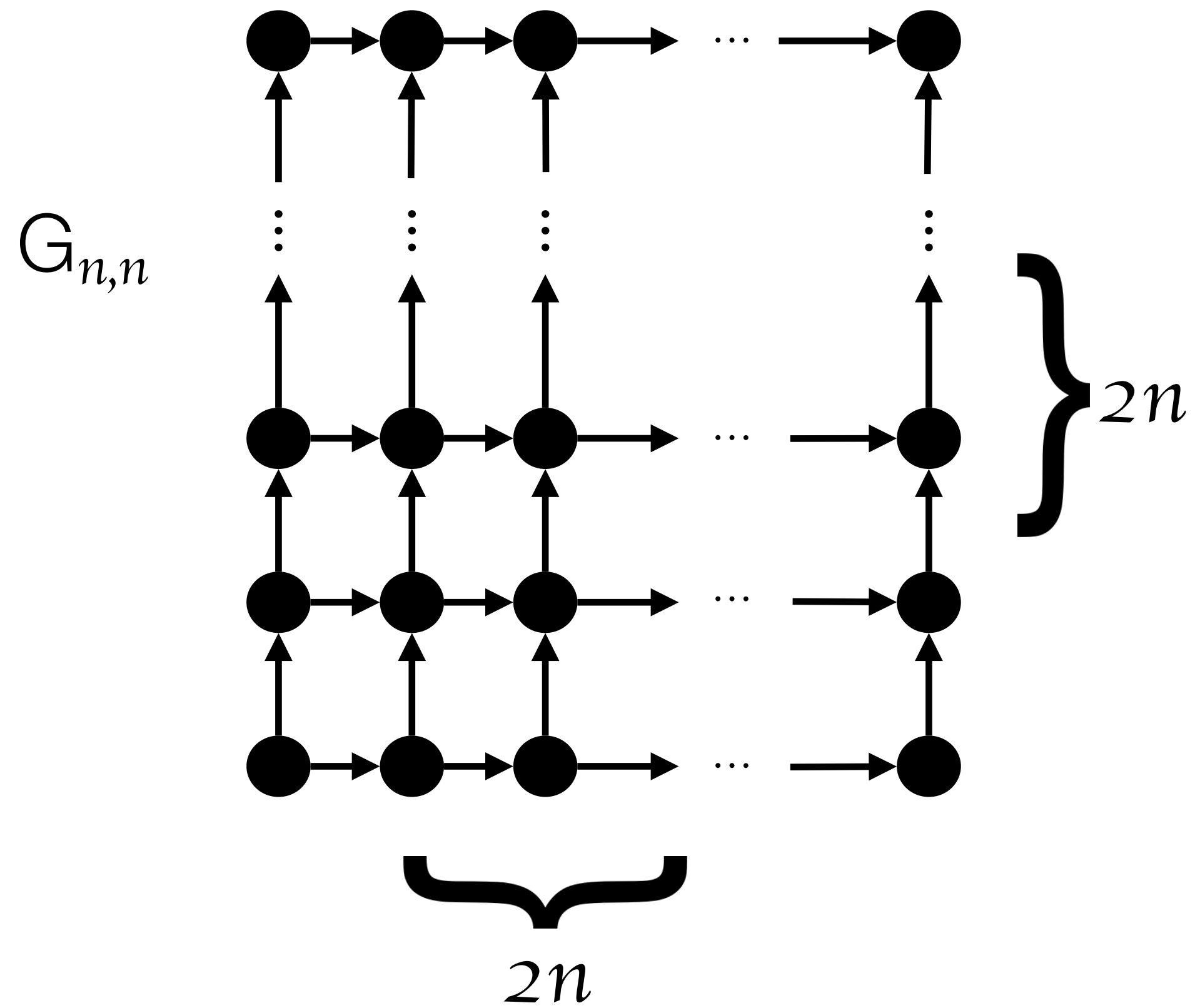
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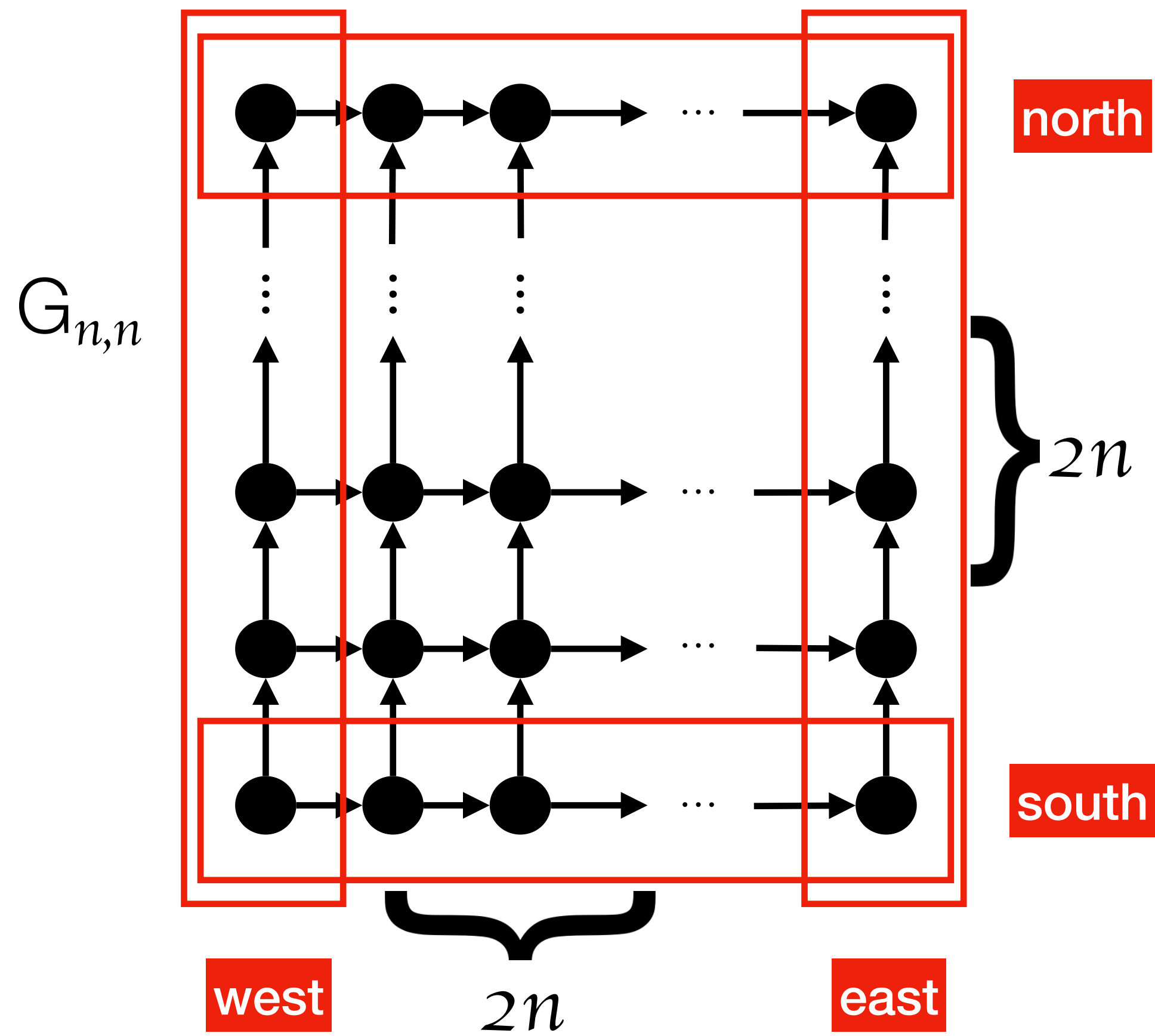
Planarity $\stackrel{\text{def}}{=}$ the graph can be drawn without crossing edges

Theorem [Kuratowski] A graph is planar iff it does not have $\mathcal{K}_5$ and $\mathcal{K}_{3,3}$ as minors

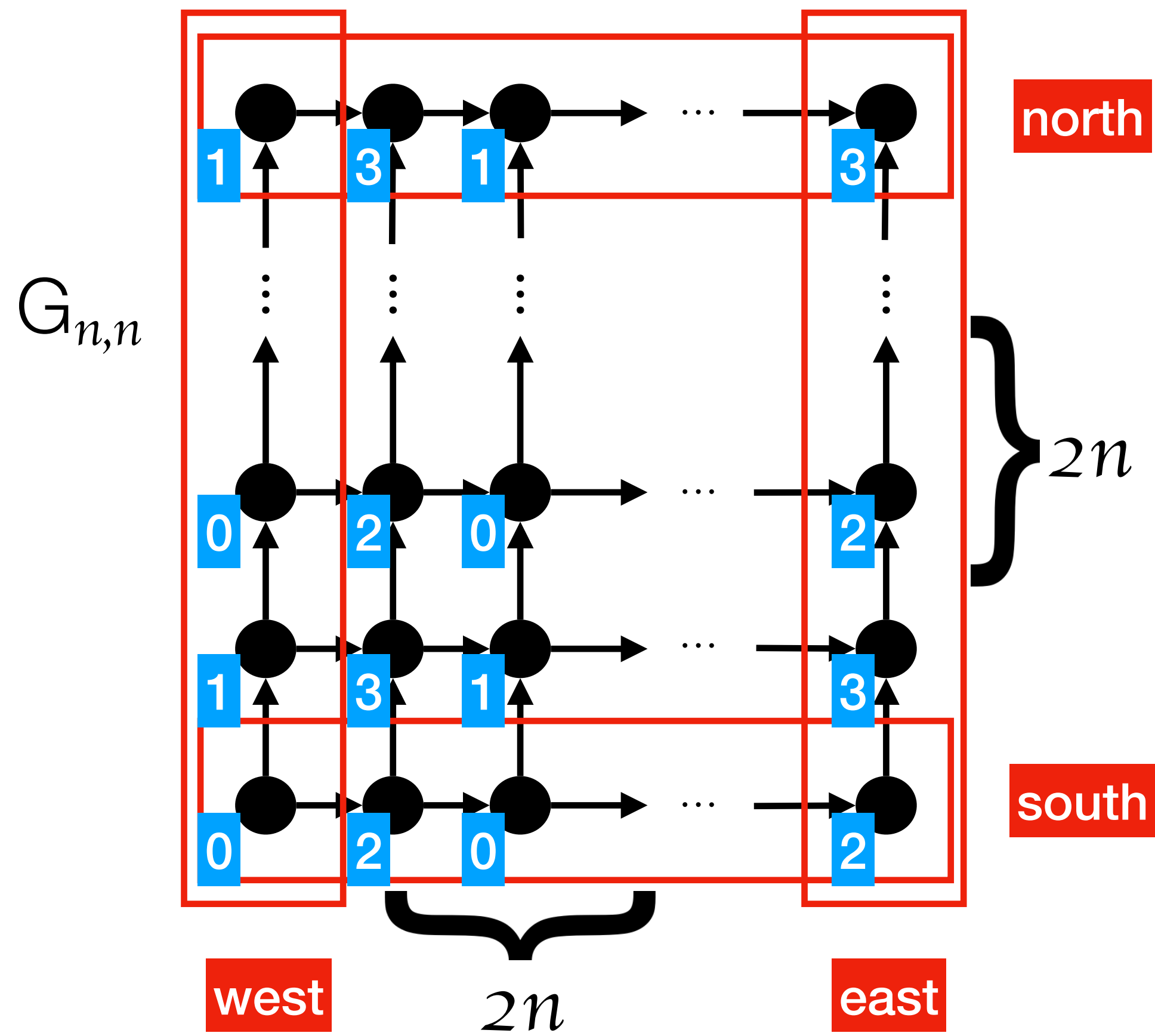
Square Grids



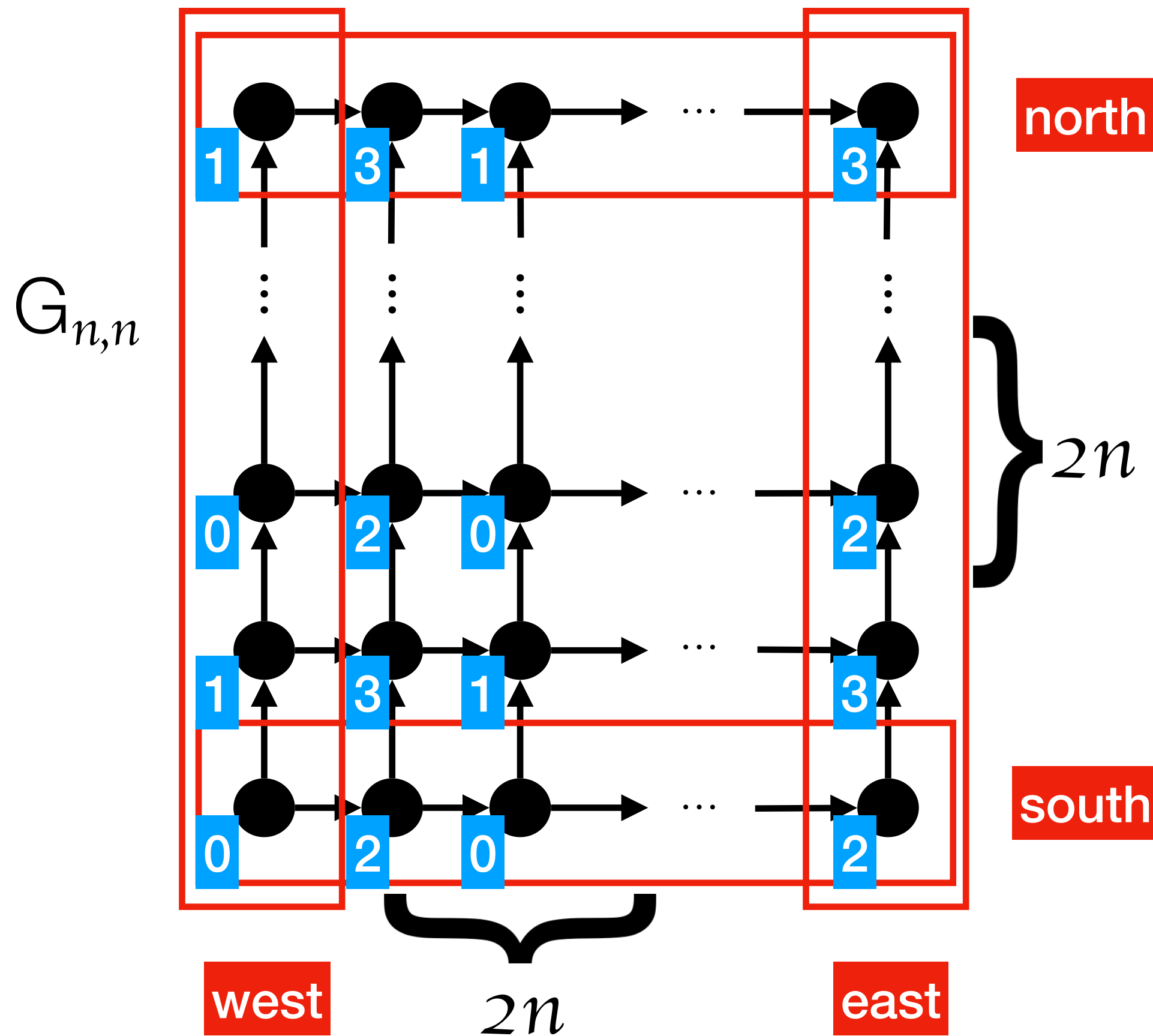
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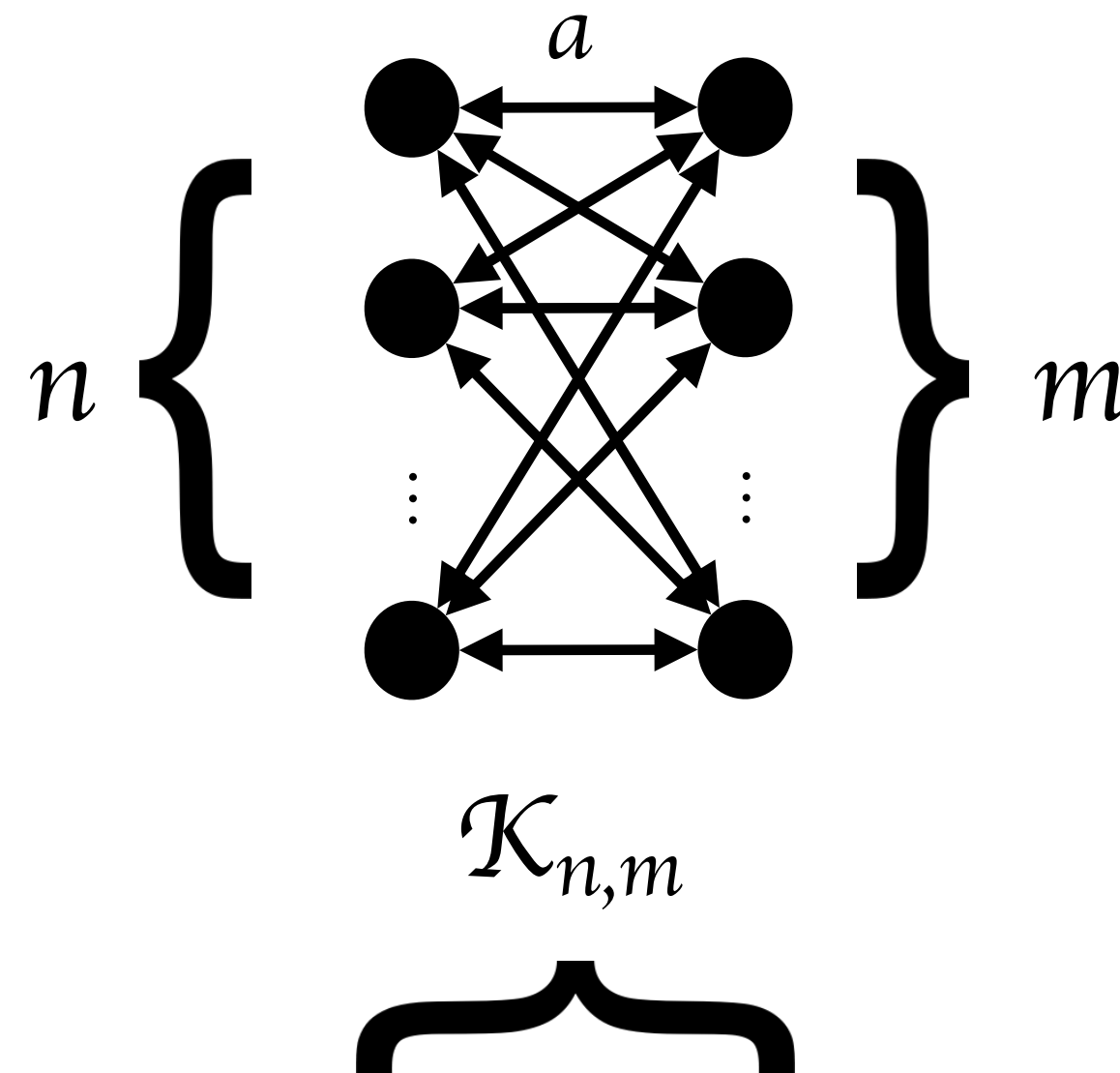


MSO₁

$\theta(\mathcal{N}, S, \mathcal{E}, \mathcal{W}, \mathcal{X}_0, \mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3)$ is the conjunction of :

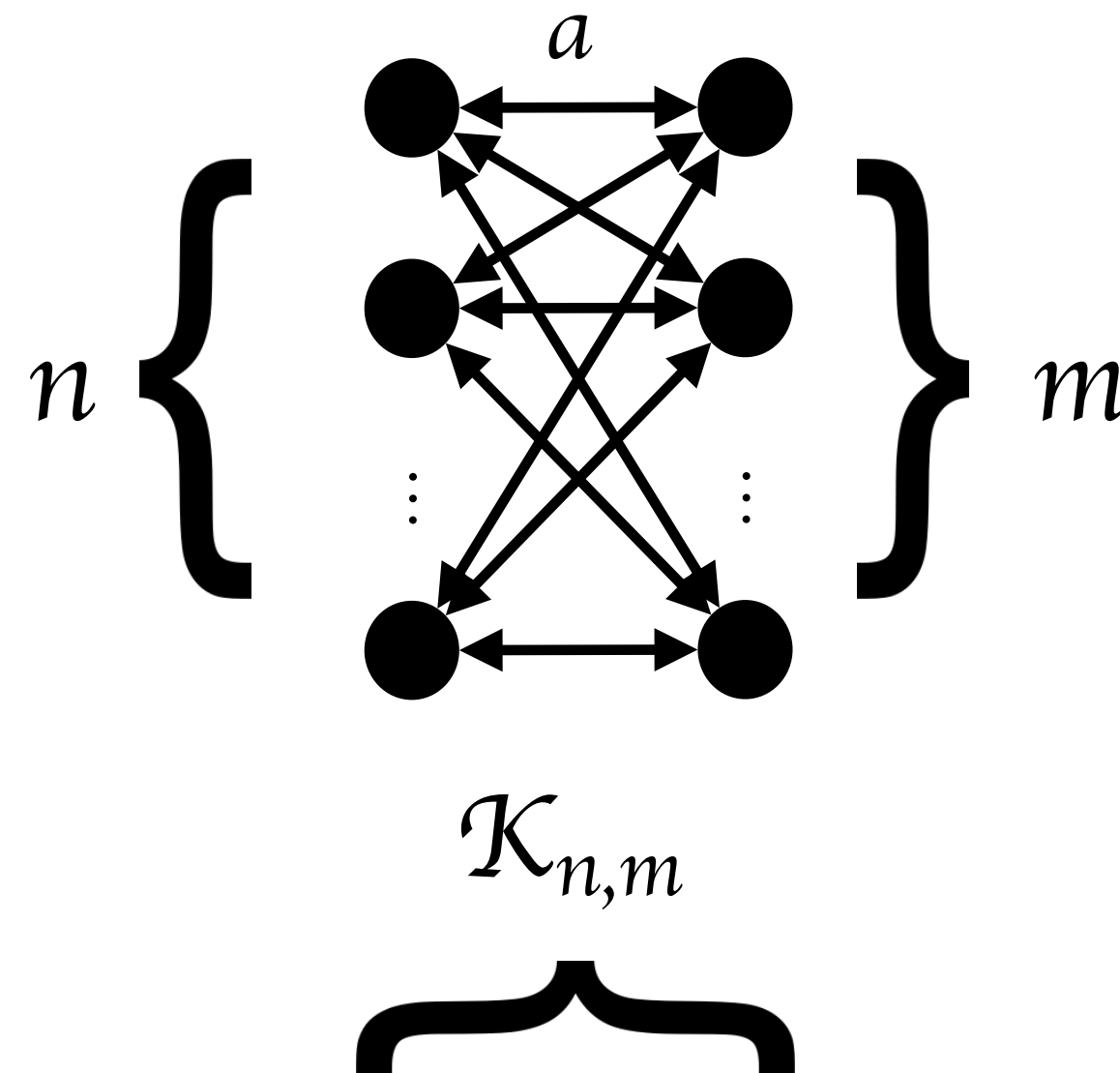
1. connected graph having no circuits
2. $\mathcal{X}_0, \mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3$ partition the vertices
3. $(\mathcal{X}_0 \cup \mathcal{X}_3, \mathcal{X}_1 \cup \mathcal{X}_2)$ -bipartite graph
4. each vertex has at most one i -predecessor (i -successor), $i=0, \dots, 3$
5. $\mathcal{N}$ is a path from $\mathcal{N} \cap \mathcal{W}$ to $\mathcal{N} \cap \mathcal{E}$ alternating between $\mathcal{X}_1$ and $\mathcal{X}_3$
6. $\mathcal{E}$ is a path from $S \cap \mathcal{E}$ to $\mathcal{N} \cap \mathcal{E}$ alternating between $\mathcal{X}_2$ and $\mathcal{X}_3$
7. similar conditions for S and $\mathcal{W}$
8. each $\mathcal{N} \cap \mathcal{X}_1$ vertex has no $\mathcal{X}_2$ -successor; each $(\mathcal{N} \setminus \mathcal{E}) \cap \mathcal{X}_3$ vertex has no $\mathcal{X}_0$ -successor; $\mathcal{N} \cap \mathcal{E}$ has no successors
9. similar conditions for $S, \mathcal{E}$ and $\mathcal{W}$
10. for each vertex $x \notin \mathcal{N} \cup \mathcal{E}$ there exist y, z, u , $y \neq z$ such that $x \rightarrow y$, $x \rightarrow z$, $y \rightarrow u$ and $z \rightarrow u$
11. the unique path from $S \cap \mathcal{W}$ alternating $\mathcal{X}_0, \mathcal{X}_2, \mathcal{X}_3, \mathcal{X}_1$ reaches $\mathcal{N} \cap \mathcal{E}$

$$\text{CMSO}_1 \not\subseteq \text{CMSO}_2$$



$$\begin{aligned} & \exists X \exists Y. (X \cap Y = \emptyset \wedge \forall x. x \in X \vee x \in Y) \wedge \\ & \forall x \forall y. (x \in X \wedge y \in Y \Rightarrow a(x, y) \wedge a(y, x)) \wedge (x \in X \wedge y \in X \Rightarrow \neg a(x, y)) \wedge (x \in Y \wedge y \in Y \Rightarrow \neg a(x, y)) \end{aligned}$$

$$\text{CMSO}_1 \subsetneq \text{CMSO}_2$$



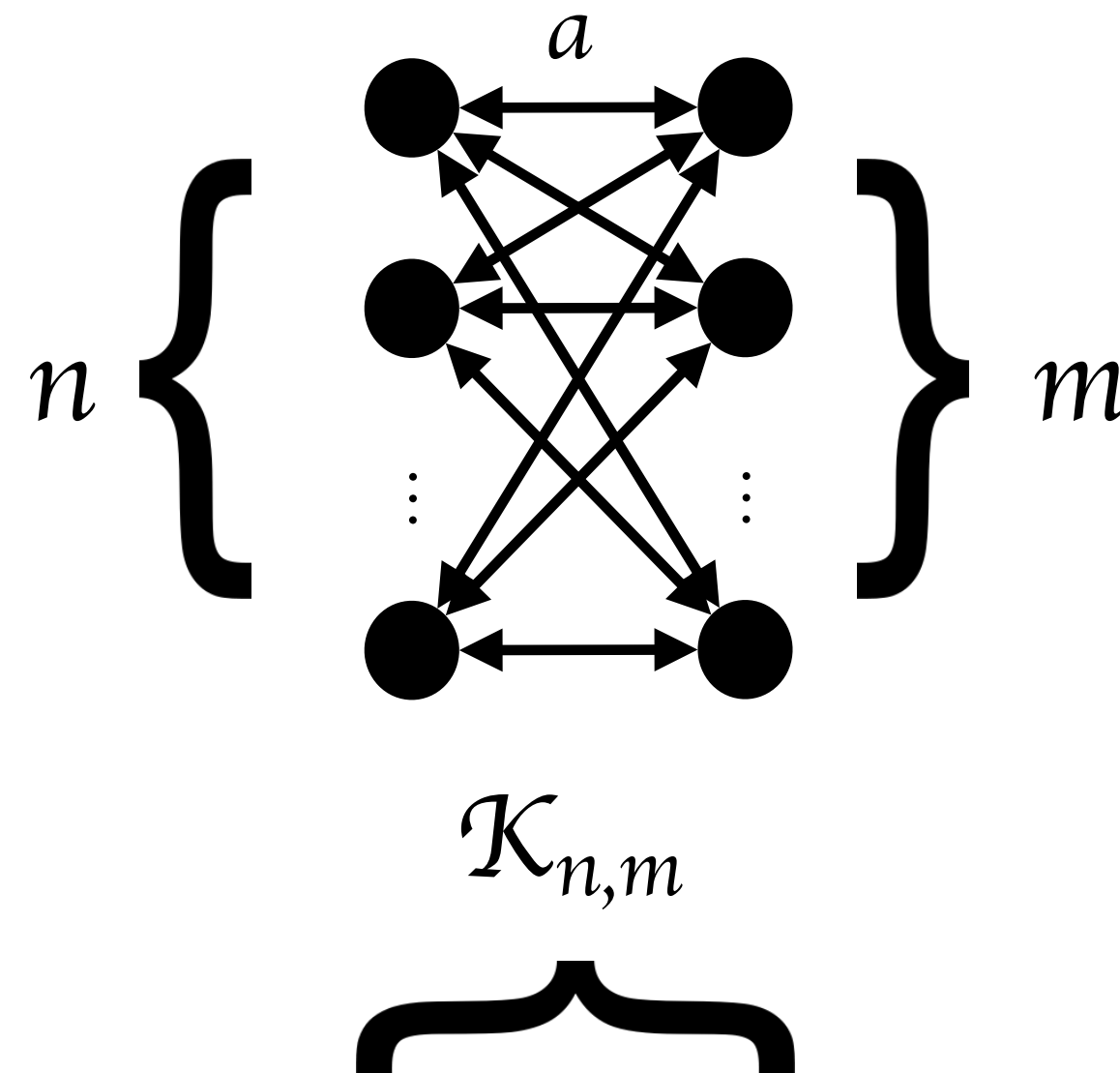
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$\mathcal{K}_{n,m}$ is Hamiltonian $\Leftrightarrow n=m$

CMSO₁ cannot express equality of sets

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MSO₁

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$\mathcal{K}_{n,m}$ is Hamiltonian $\Leftrightarrow n=m$ CMSO₁ cannot express equality of sets

Theorem [Sparseness] Let $k \geq 1$ and $\mathcal{L}$ be a set of graphs such that $|E_G| \leq k |V_G|$, for all $G \in \mathcal{L}$. Every CMSO₂ sentence ψ can be translated into a CMSO₁ sentence ψ' such that $|G| \models \psi \Leftrightarrow |G| \models \psi'$

Hypergraphs

$\Sigma \stackrel{\text{def}}{=} \text{finite ranked alphabet of edge labels}$

$G = (V, E, \lambda, v, s) \stackrel{\text{def}}{=} \text{hypergraph}$

► $V \stackrel{\text{def}}{=} \text{finite set of vertices}$

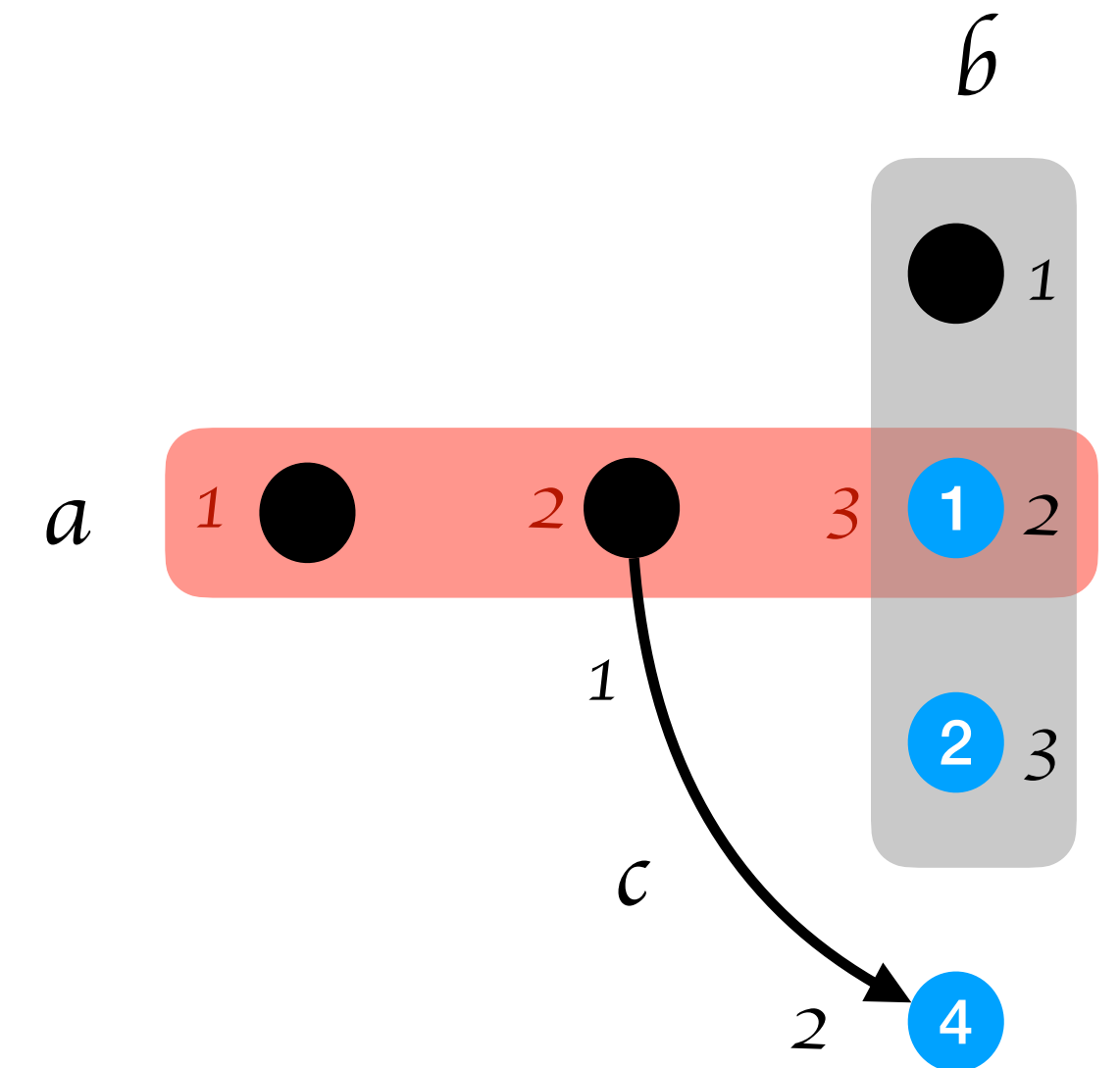
► $E \stackrel{\text{def}}{=} \text{finite set of edges}$

► $\lambda : E \rightarrow \Sigma \stackrel{\text{def}}{=} \text{edge labeling}$

► $v : E \rightarrow V^+ \stackrel{\text{def}}{=} \text{incidence relation; } \#\lambda(e) = |v(e)| \text{ for each } e \in E$

► $s : \mathbb{N} \rightarrow V \stackrel{\text{def}}{=} \text{finite injective partial source labeling}$

The **sort** of $G \stackrel{\text{def}}{=} \text{dom}(s)$



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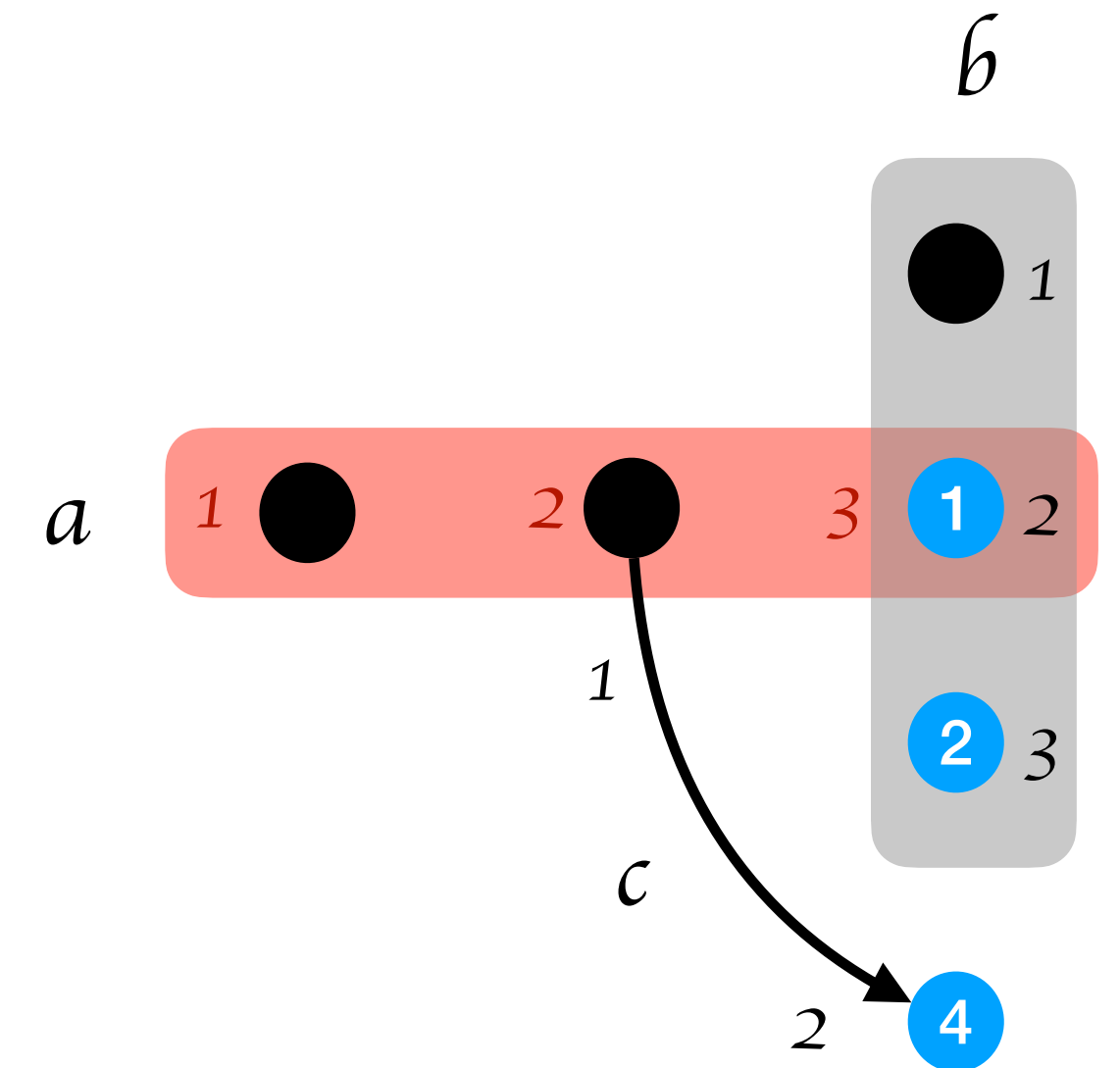
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The **sort** of $G \stackrel{\text{def}}{=} \text{dom}(s)$

$\mathcal{G}(\Sigma)_\tau \stackrel{\text{def}}{=} \text{set of hypergraphs of sort } \tau \subseteq_{\text{fin}} \mathbb{N} \text{ with edge labels from } \Sigma$

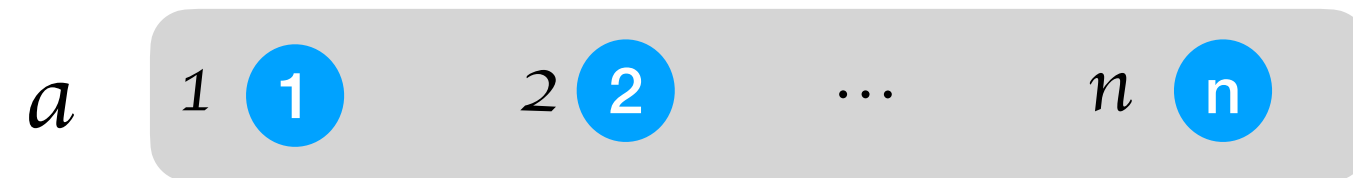
$\mathcal{G}(\Sigma, S) \stackrel{\text{def}}{=} \cup \{ \mathcal{G}(\Sigma)_\tau \mid \tau \subseteq S \}$

$\mathcal{G}(\Sigma) \stackrel{\text{def}}{=} \text{set of hypergraphs of any sort}$



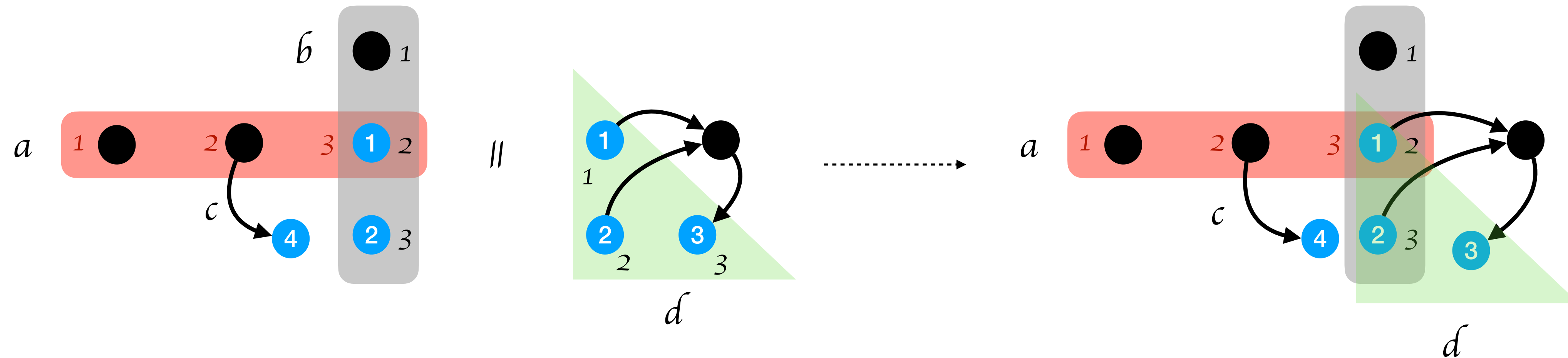
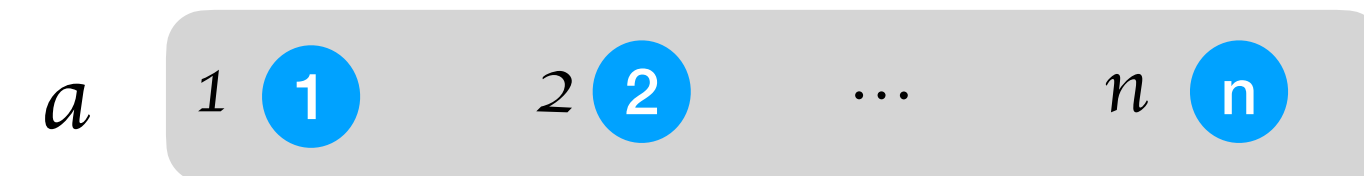
Hyperedge-Replacement (HR) Algebra

For each $a \in \Sigma$, $\#a=n$



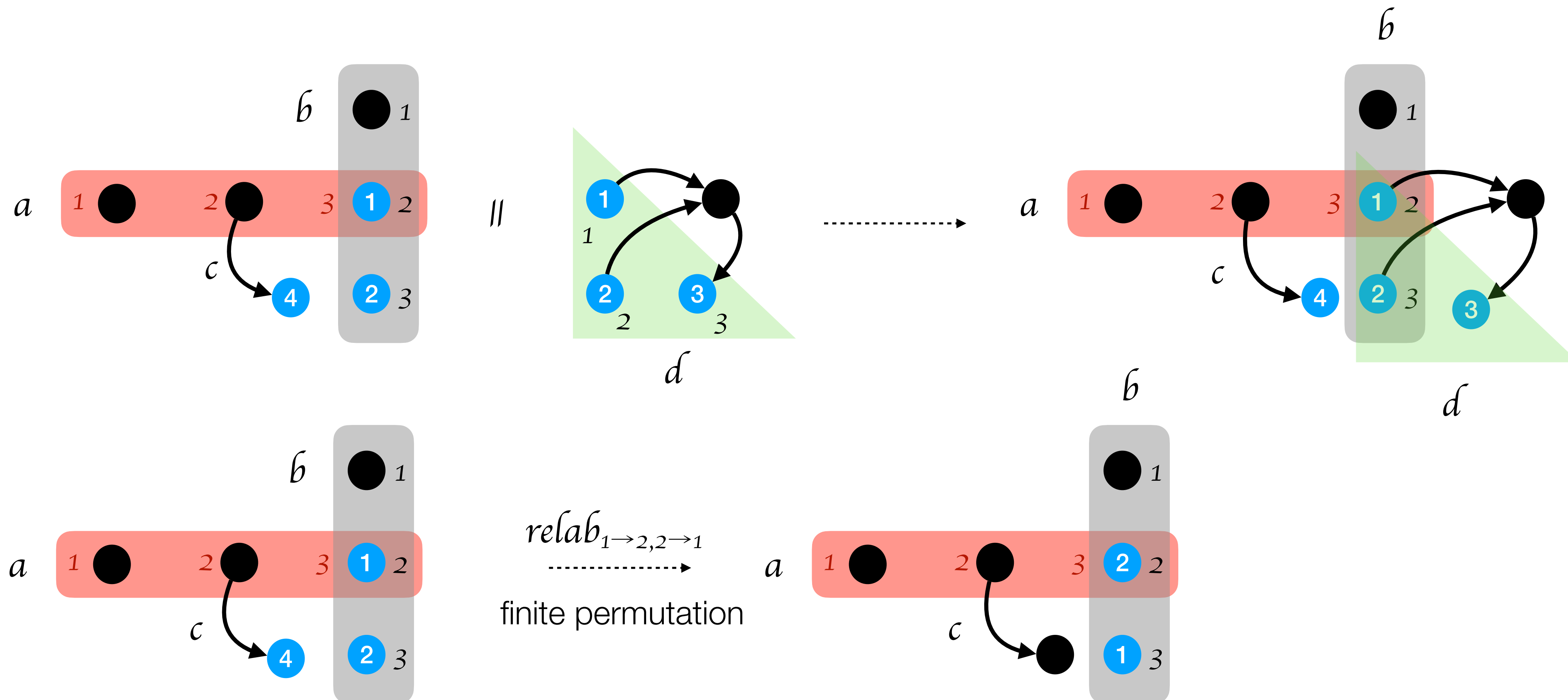
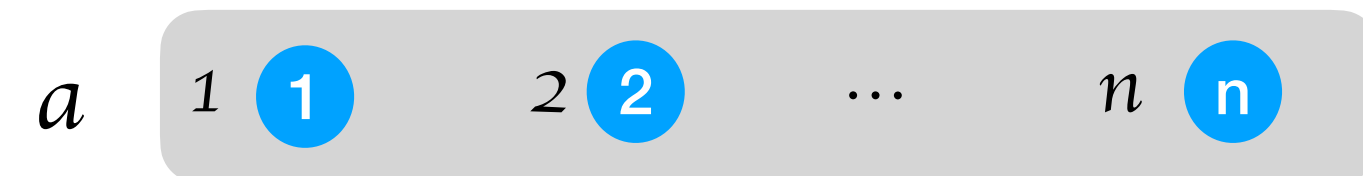
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Recognisability in the HR algebra

$$\begin{array}{ccc} (G_{\tau}(\Sigma), \parallel^{\mathcal{HR}}, \{relab_{\alpha}^{\mathcal{HR}}\}_{\alpha : \mathbb{N} \rightarrow \mathbb{N}}, \{a^{\mathcal{HR}}\}_{a \in \Sigma}) & \xrightarrow{\quad h \quad} & (A_{\tau}, \parallel^{\mathcal{A}}, \{relab_{\alpha}^{\mathcal{A}}\}_{\alpha : \mathbb{N} \rightarrow \mathbb{N}}, \{a^{\mathcal{A}}\}_{a \in \Sigma}) \\ (G_{\tau}(\Sigma), \parallel^{\mathcal{HR}}, \{relab_{\alpha}^{\mathcal{HR}}\}_{\alpha : \mathbb{N} \rightarrow \mathbb{N}}, \{a^{\mathcal{HR}}\}_{a \in \Sigma}) & \xrightarrow{\quad h \quad} & (A_{\tau}, \parallel^{\mathcal{A}}, \{relab_{\alpha}^{\mathcal{A}}\}_{\alpha : \mathbb{N} \rightarrow \mathbb{N}}, \{a^{\mathcal{A}}\}_{a \in \Sigma}) \\ (G_{\tau}(\Sigma), \parallel^{\mathcal{HR}}, \{relab_{\alpha}^{\mathcal{HR}}\}_{\alpha : \mathbb{N} \rightarrow \mathbb{N}}, \{a^{\mathcal{HR}}\}_{a \in \Sigma}) & \xrightarrow{\quad h \quad} & (A_{\tau}, \parallel^{\mathcal{A}}, \{relab_{\alpha}^{\mathcal{A}}\}_{\alpha : \mathbb{N} \rightarrow \mathbb{N}}, \{a^{\mathcal{A}}\}_{a \in \Sigma}) \end{array}$$

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 \end{array}$$

$\sqcup$ recognisable
 $\sqcup$ locally finite

$= h^{-1}(F)$

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 \end{array}$$

Why locally finite and not just finite ?

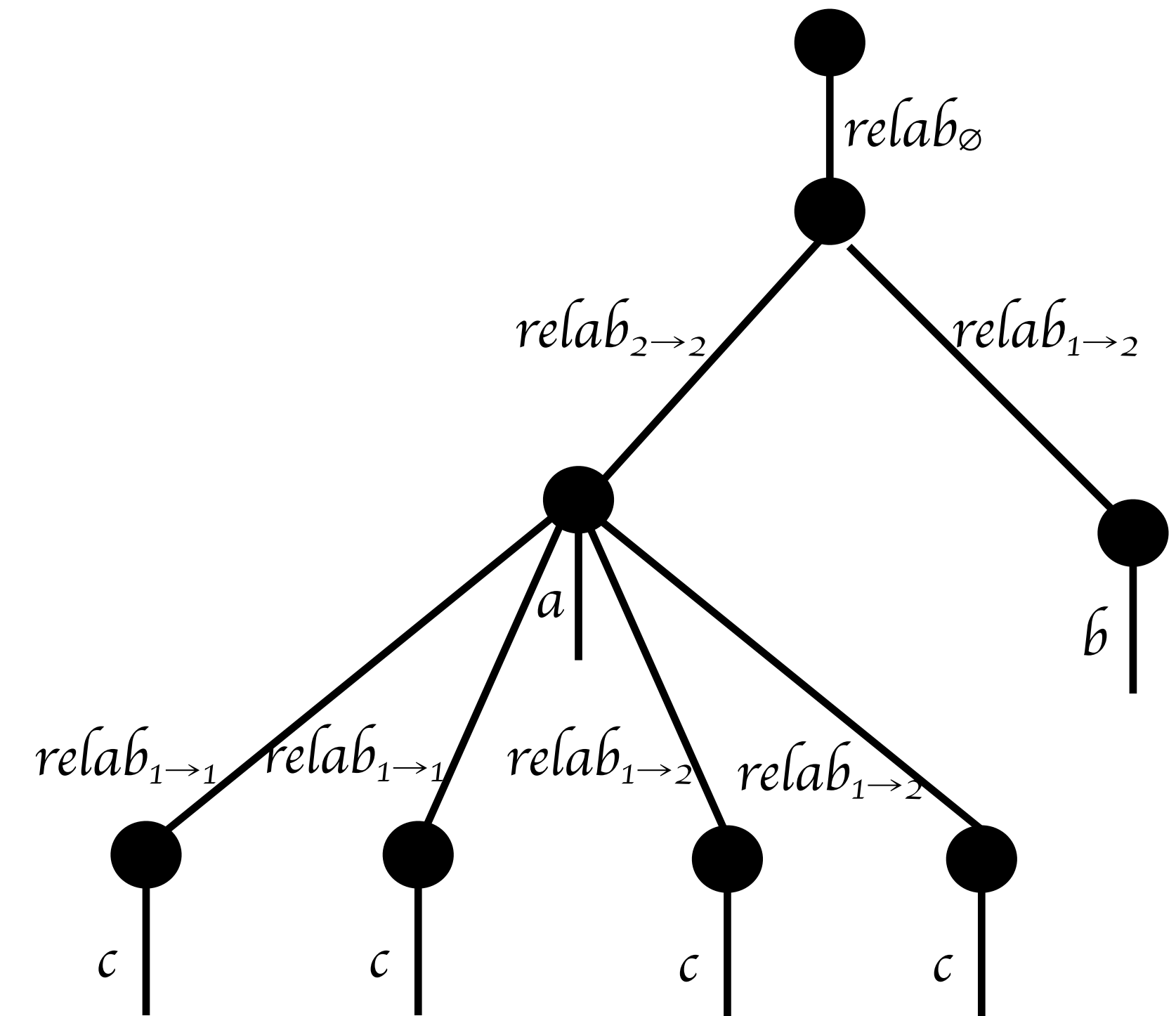
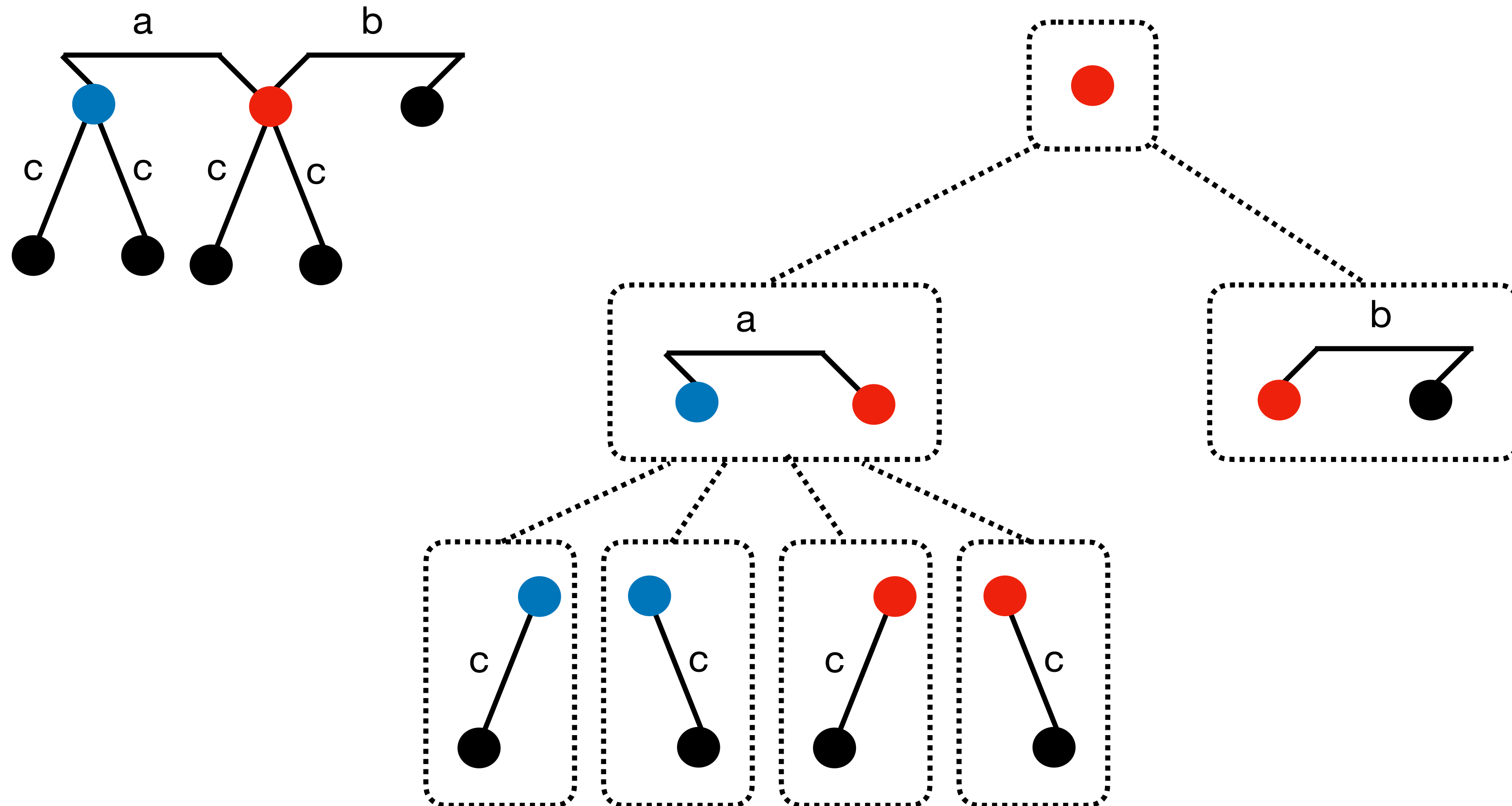
Square grids are MSO_1 -definable but recognising the set $\{G_{n,n} \mid n \in \mathbb{N}\}$ requires infinitely many sorts

Tree-width (algebraic)

$\mathcal{T}(\mathcal{HR}) \stackrel{\text{def}}{=} \text{set of } \mathcal{HR}\text{-terms}$

$tw(G) \stackrel{\text{def}}{=} \min\{ (\text{number of source labels occurring in } t) - 1 \mid t \in \mathcal{T}(\mathcal{HR}), t^{\mathcal{HR}} = G \}$

$\mathcal{G}^{tw \leq k}(\Sigma) \stackrel{\text{def}}{=} \text{set of graphs of tree-width at most } k \text{ with labels in } \Sigma$



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CMSO₂-definability $\Rightarrow$ Recognisability in HR

Theorem [Courcelle] Let Σ be finite alphabet of edge labels.

Each language $\mathcal{L} \subseteq \mathcal{G}(\Sigma)$ is CMSO₂-definable $\Rightarrow \mathcal{L}$ is recognisable in HR

$X = \emptyset$	$X \subseteq Y$	$s_i(X)$	$edg_a(Y, X_1, \dots, X_n)$	$ X = p \bmod q$
$single(X)$	X is a singleton and	X is a singleton and	$Y, X_1, \dots, X_n$ are singletons	
X is a singleton	$x \in X$ is the i -th source	$x \in X$ is the i -th source	$y \in Y$ edge attached to $x_1 \in X_1, \dots, x_n \in X_n$	

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 for some effectively computable boolean combination β
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Lemma 2 $relab_\alpha(G) \models \varphi(X_1, \dots, X_n) \Leftrightarrow \beta(G \models \psi_{11}, \dots, G \models \psi_{1k})$
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Let $\mathcal{L} = \mathcal{L}(\varphi)$, for a CMSO₂ sentence φ such that $qr(\varphi) = k$

$\mathcal{F}_{\leq k} \stackrel{\text{def}}{=} \text{the finite set of sentences of quantifier rank } \leq k$

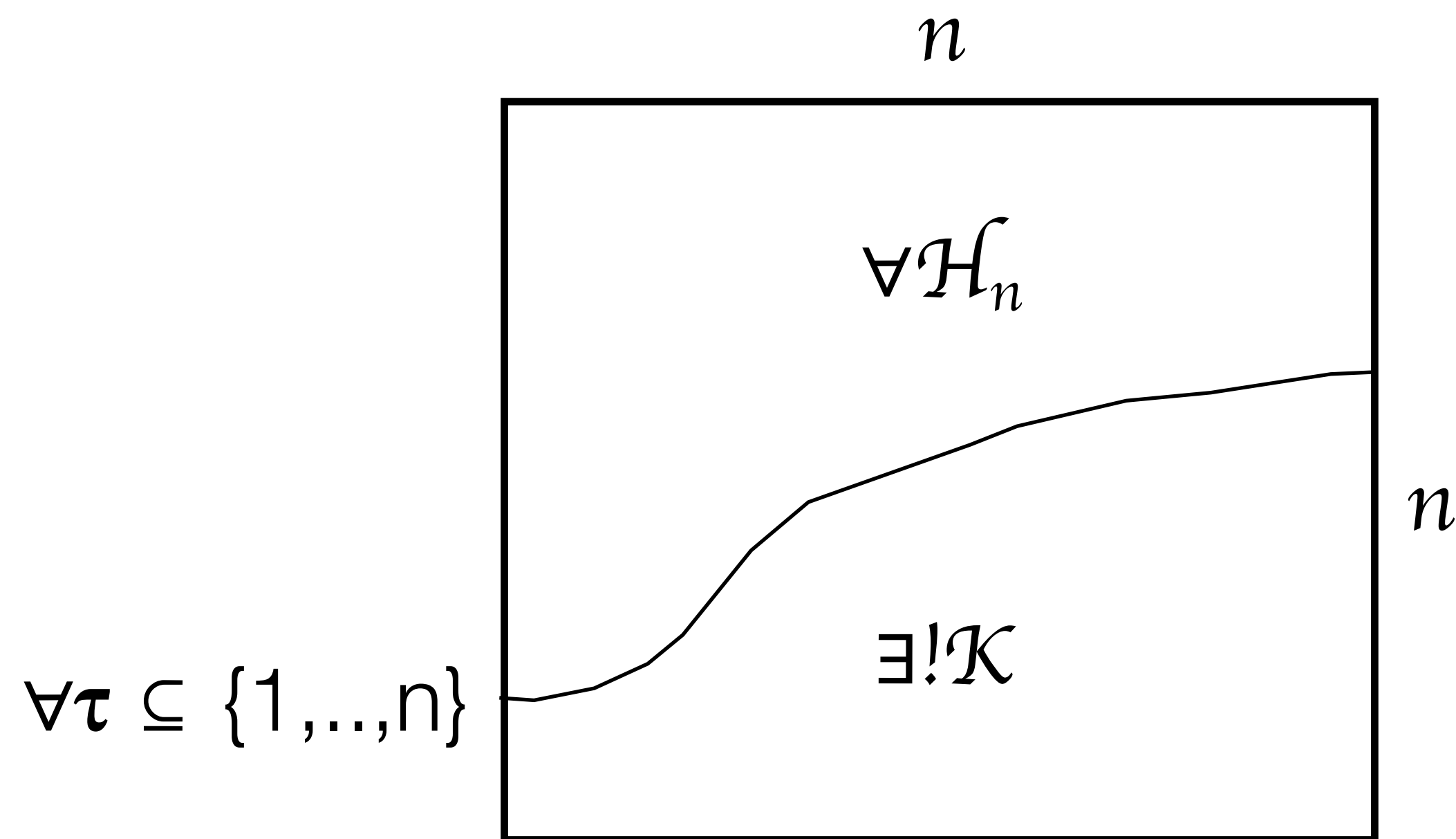
$$\theta_\tau \stackrel{\text{def}}{=} \bigwedge_{i \in \tau} \exists x . s_i(x) \wedge \bigwedge_{\neg i \in \tau} \neg \exists x . s_i(x)$$

$$\mathcal{F}_{k,\tau} \stackrel{\text{def}}{=} \{\psi \wedge \theta_\tau \mid \psi \in \mathcal{F}_{\leq k}\}$$

$$\begin{array}{ccc} (\{\mathcal{G}_\tau(\Sigma)\}_{\tau \subseteq_{\text{fin}} \mathbb{N}}, \parallel^{\mathcal{G}(\Sigma)}, relab_\alpha^{\mathcal{G}(\Sigma)}, \{a^{\mathcal{G}(\Sigma)}\}_{a \in \Sigma}) & \xrightarrow{h(G) \stackrel{\text{def}}{=} \{\psi \in \mathcal{F}_{\leq k} \mid G \models \psi\}} & (\{\mathcal{F}_{k,\tau}\}_{\tau \subseteq_{\text{fin}} \mathbb{N}}, \parallel^{\mathcal{F}_k}, relab_\alpha^{\mathcal{F}_k}, \{a^{\mathcal{F}_k}\}_{a \in \Sigma}) \\ \text{IU} & & \text{IU} \\ \mathcal{L} & = & h^{-1}(\{\varphi\}) \end{array}$$

CMO₂-definability $\not\equiv$ Recognisability in HR

Theorem [Courcelle] For each set $\mathcal{N} \subseteq \mathbb{N}$, the set $\{G_{n,n} \mid n \in \mathcal{N}\}$ is recognizable in HR



There are uncountably many sets $\mathcal{N} \subseteq \mathbb{N}$ but only countably many CMO₂ formulae

HR-Context-Free Sets of Graphs

$\mathcal{L} \subseteq \mathcal{G}(\Sigma)$ is HR-context-free $\stackrel{\text{def}}{=} \mathcal{L} = \{t^{\mathcal{G}(\Sigma)} \mid t \in \mathcal{T}\}$, $\mathcal{T}$ recognisable set of HR terms

$\mathcal{L} = \mathcal{L}^{\mathcal{G}(\Sigma)}(\Gamma)$, for some grammar $\Gamma = (\mathbb{U}, \leftarrow)$ having rules of the forms :

$$\mathcal{U} \leftarrow \mathcal{U}_1 \parallel \mathcal{U}_2$$

$$\mathcal{U} \leftarrow \text{relab}_\alpha(\mathcal{U}_1), \alpha \text{ finite permutation on } \mathbb{N}$$

$$\mathcal{U} \leftarrow a, a \in \Sigma$$

$$\leftarrow \mathcal{U}$$

CMSO₂ over Tree-width Bounded Graphs

Theorem [Courcelle] Let Σ be finite alphabet of edge labels.

For each $k \geq 1$, CMSO₂ has a decidable satisfiability problem over $\mathcal{L}^{\leq k} \stackrel{\text{def}}{=} \{G \in \mathcal{G}(\Sigma) \mid \text{tw}(G) \leq k\}$

The following grammar $\Gamma_{\leq k}$ produces all graphs of tree-width k at most:

$$\mathcal{U} \leftarrow \mathcal{U} \parallel \mathcal{U}$$

$$\mathcal{U} \leftarrow \text{relab}_{\alpha}(\mathcal{U}), \text{ for each permutation } \alpha \text{ of a subset of } \{1, 2, \dots, k\}$$

$$\mathcal{U} \leftarrow a, \text{ for each } a \in \Sigma, \#a \leq k$$

$$\leftarrow \mathcal{U}$$

For a CMSO₂ formula φ , build a grammar $\Gamma^{qr(\varphi)}_{\leq k}$ such that $\mathcal{L}(\Gamma^{qr(\varphi)}_{\leq k}) = \mathcal{L}(\Gamma_{\leq k}) \cap h^{-1}(\{\varphi\})$ **(FT)**

Check if $\mathcal{L}(\Gamma^{qr(\varphi)}_{\leq k}) = \emptyset$

Relational Structures

$$\mathcal{R} = r_1, r_2, \dots$$

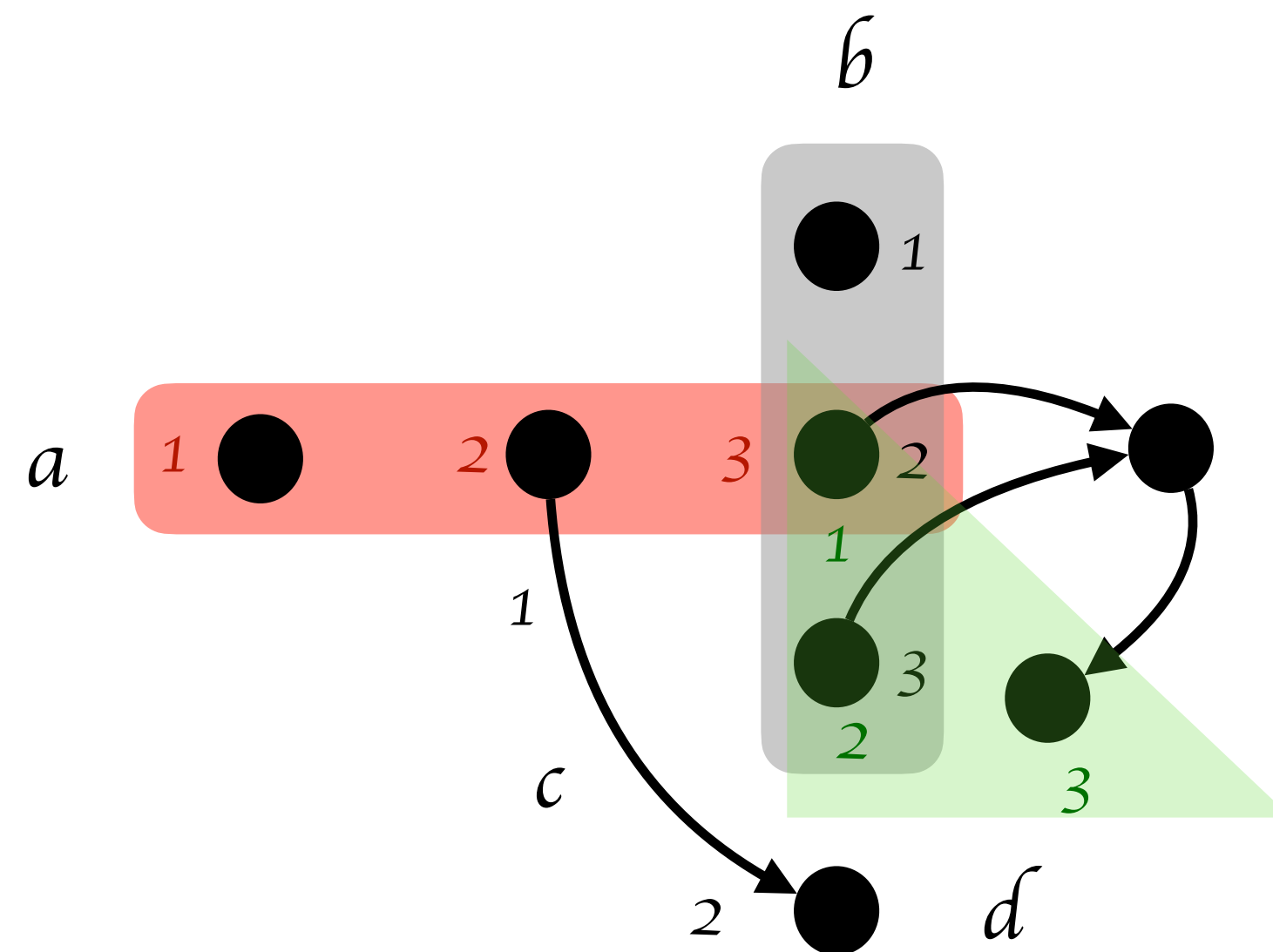
signature of relation symbols

$$S = (\mathcal{U}, r^S_1, r^S_2, \dots)$$

relational structure

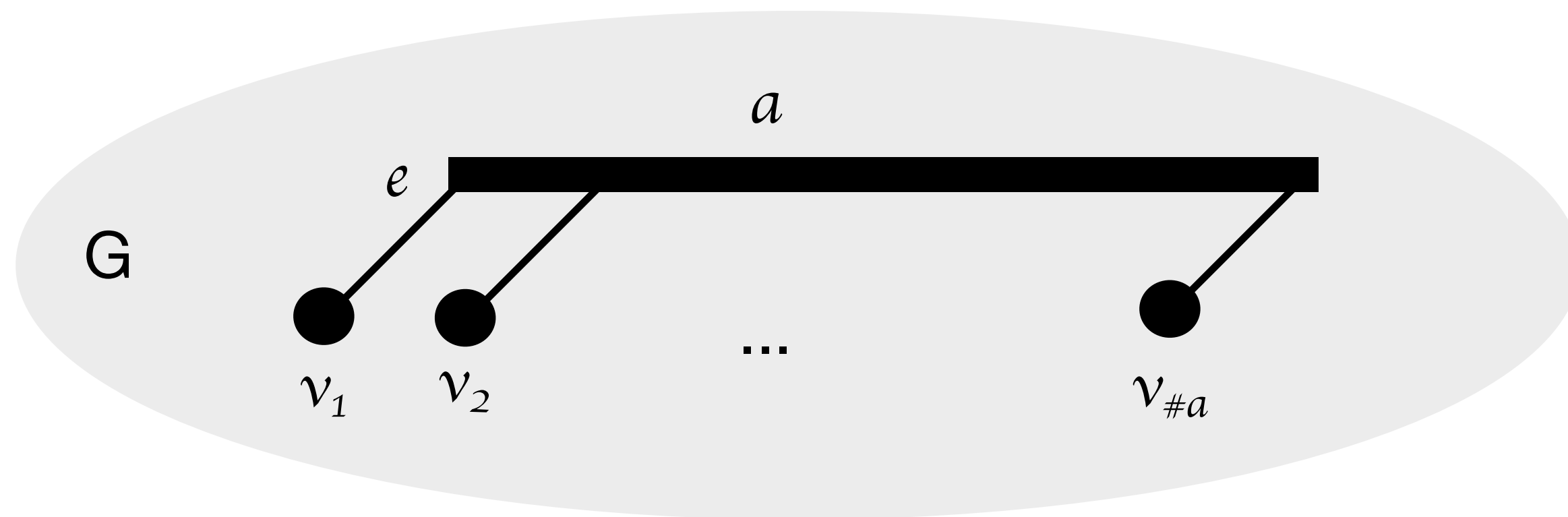
universe

interpretation of
relation symbols

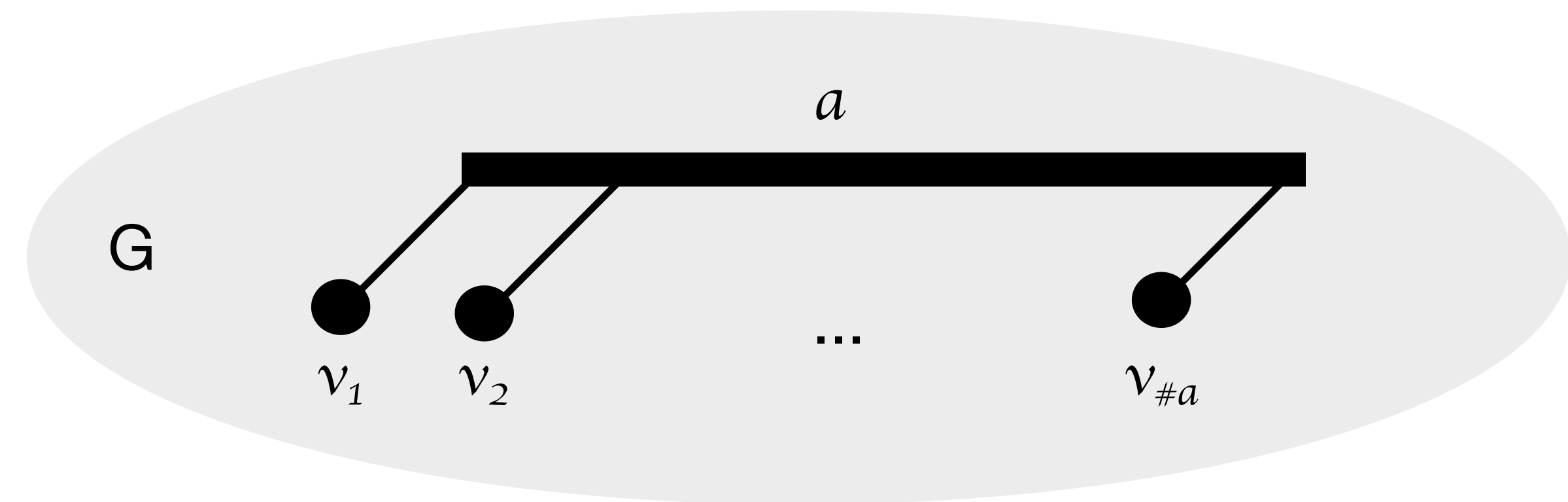


Hypergraphs as Relational Structures

Incidence encoding (CMSO₂)



Adjacency encoding (CMSO₁)



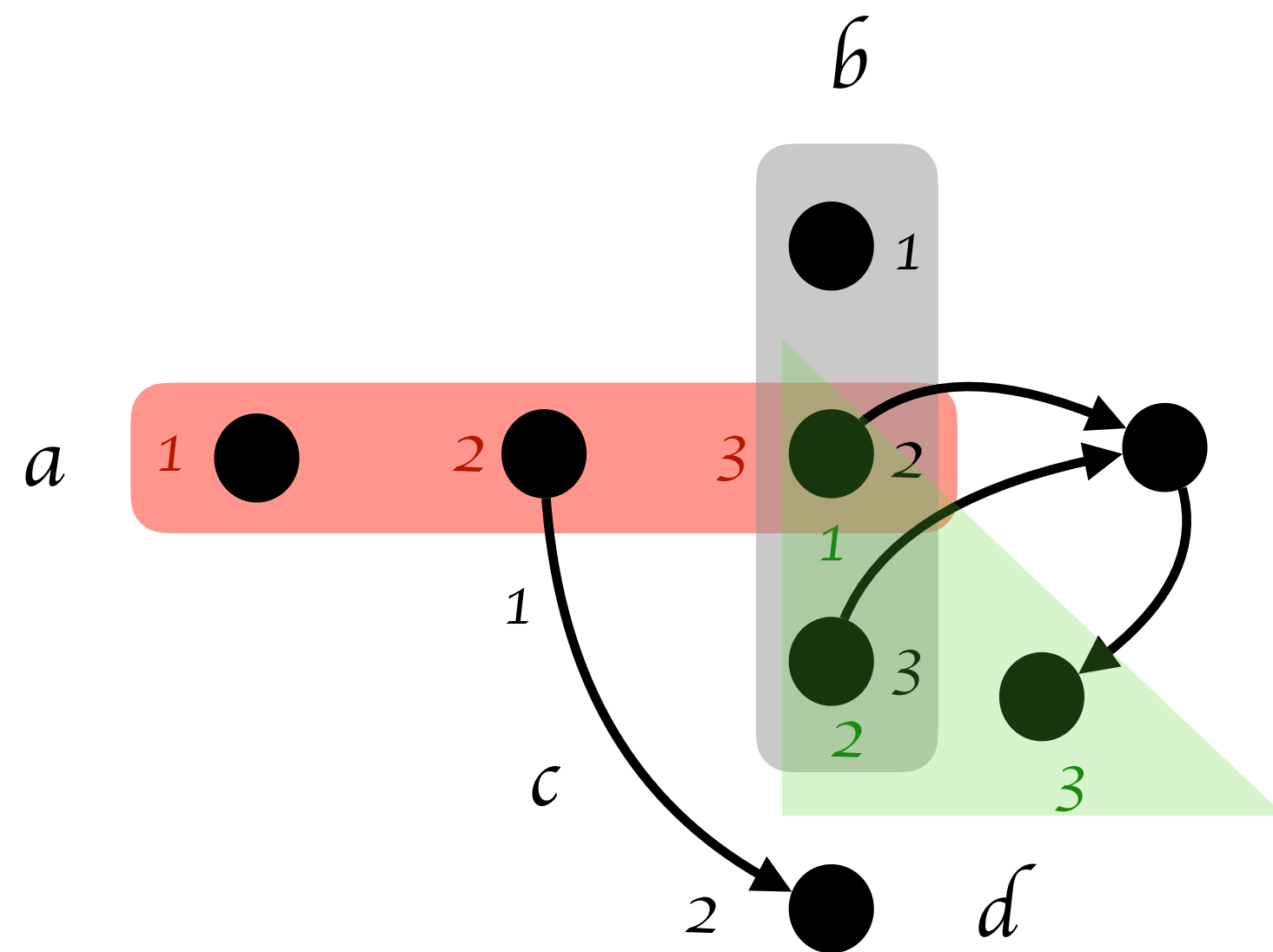
$\mathcal{U}_{\|G\|} \stackrel{\text{def}}{=} V_G \cup E_G$ (both vertices and edges)

$edg_a^{\|G\|} \stackrel{\text{def}}{=} \{(e, v_1, \dots, v_{\#a}) \mid e \in E_G, \lambda_G(e)=a, v_G(e)=(v_1, \dots, v_{\#a})\}$

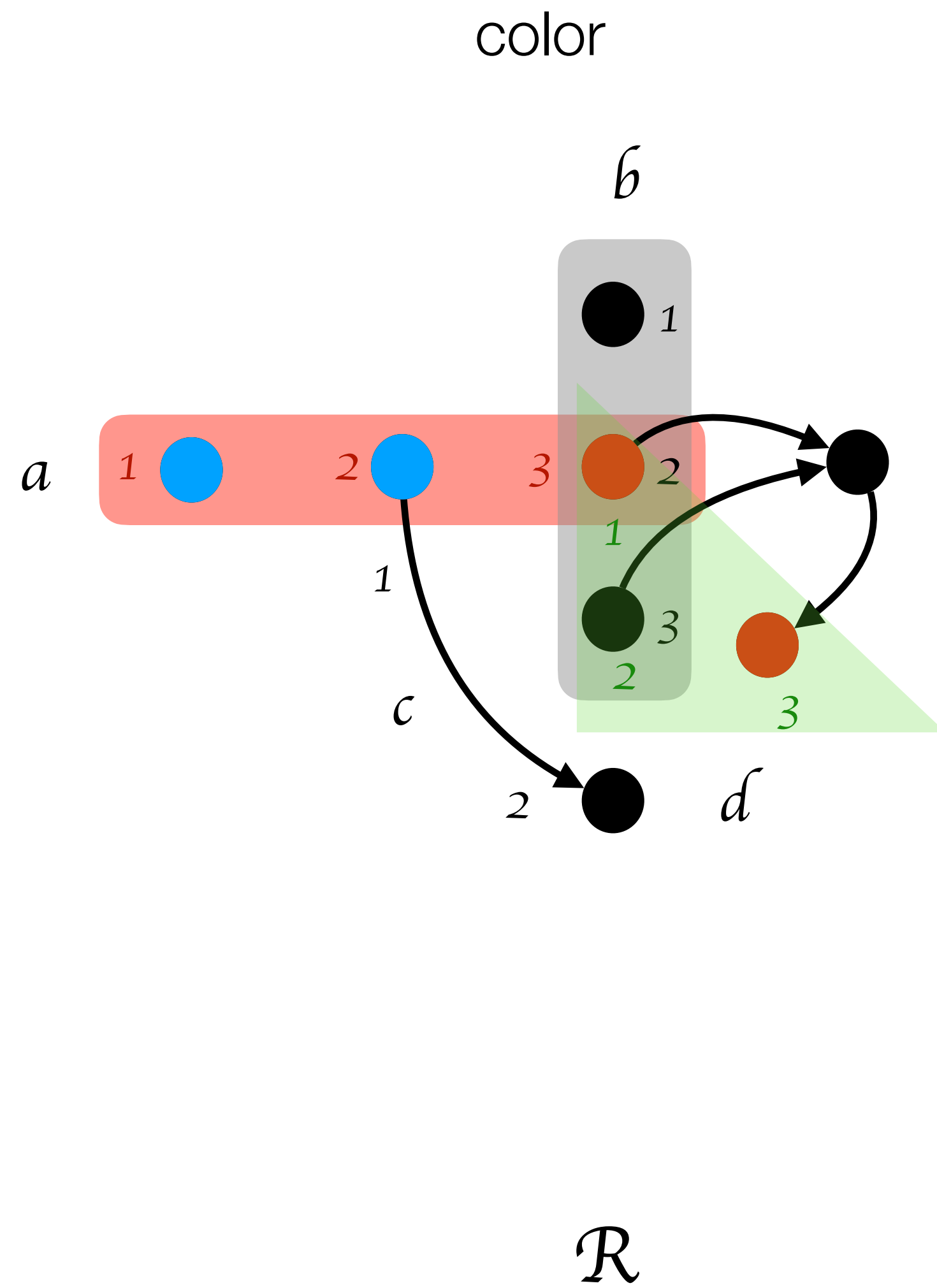
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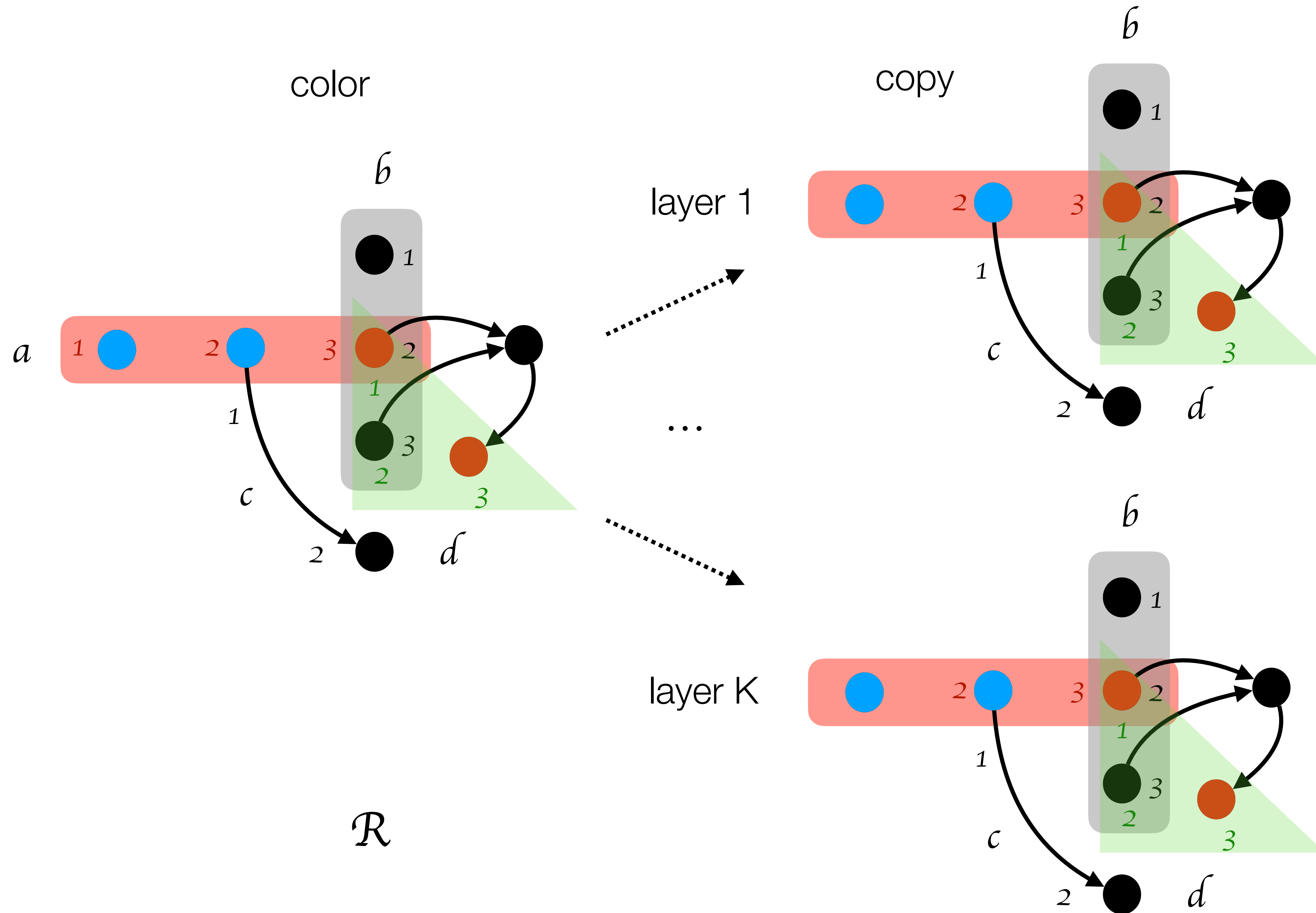
Definable Transductions



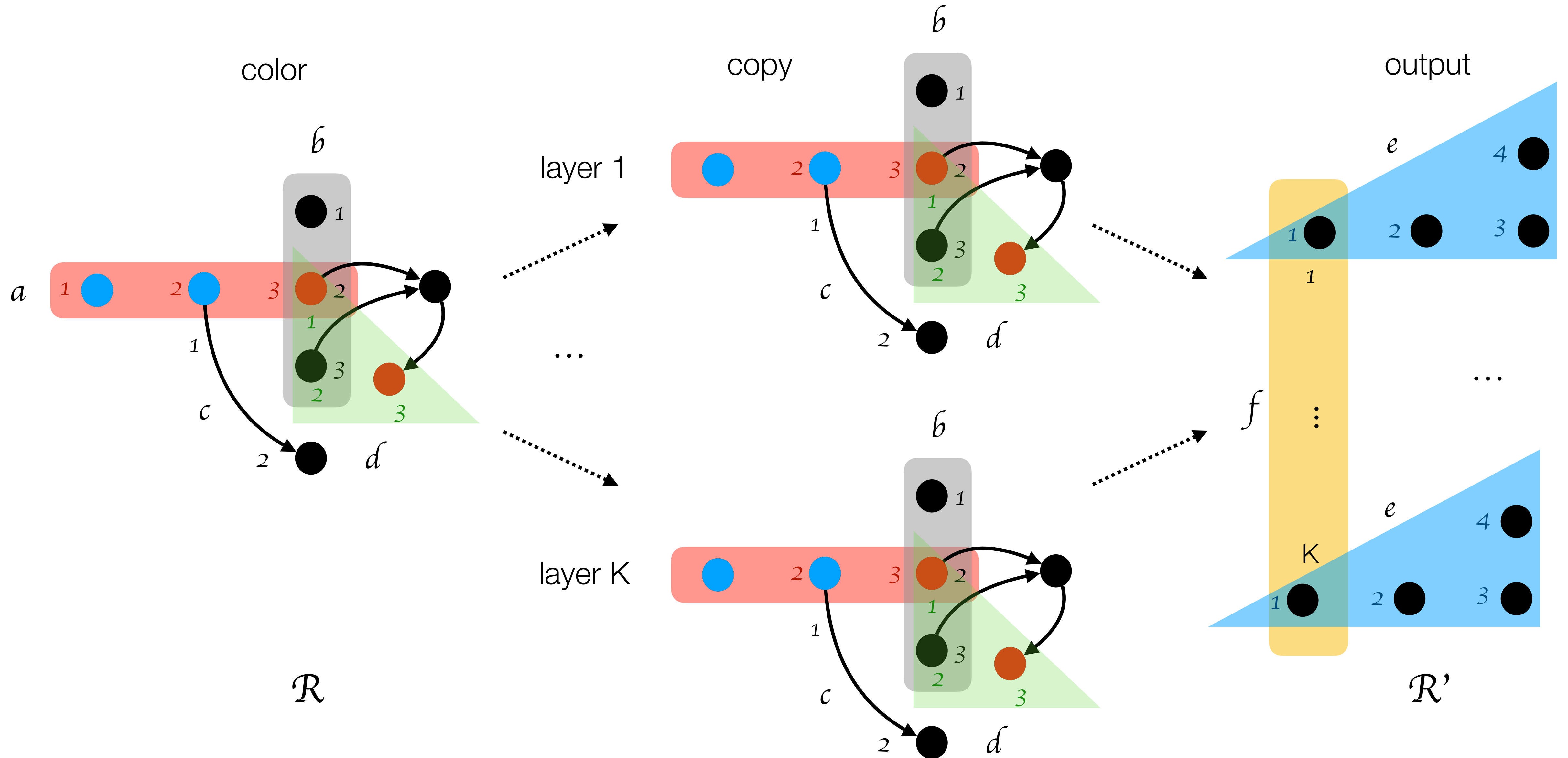
Definable Transductions



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Definable Transductions

$\mathcal{R}$ - $\mathcal{R}'$ -definition scheme

MSO formulae using the signature $\mathcal{R}$



$$\Delta = \left(\underbrace{\varphi(X_1, \dots, X_n)}_{\text{coloring}}, \underbrace{\{\psi_i(x_1, X_1, \dots, X_n)\}_{1 \leq i \leq K}}_{i\text{-layer domain}}, \underbrace{\{\theta_{r, i_1, \dots, i_{\#r}}(x_1, \dots, x_{\#r}, X_1, \dots, X_n)\}_{r \in \mathcal{R}', 1 \leq i_1, \dots, i_{\#r} \leq K}}_{\text{relation } r \in \mathcal{R}'} \right)$$



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i -layer domain



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Δ defines a relation δ from $\mathcal{R}$ -structures to $\mathcal{R}'$ -structures

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The pre-image $\delta^{-1}(S)$ of a definable set S of $\mathcal{R}'$ -structures is a definable set of $\mathcal{R}$ -structures

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MSO formulae using the signature $\mathcal{R}$

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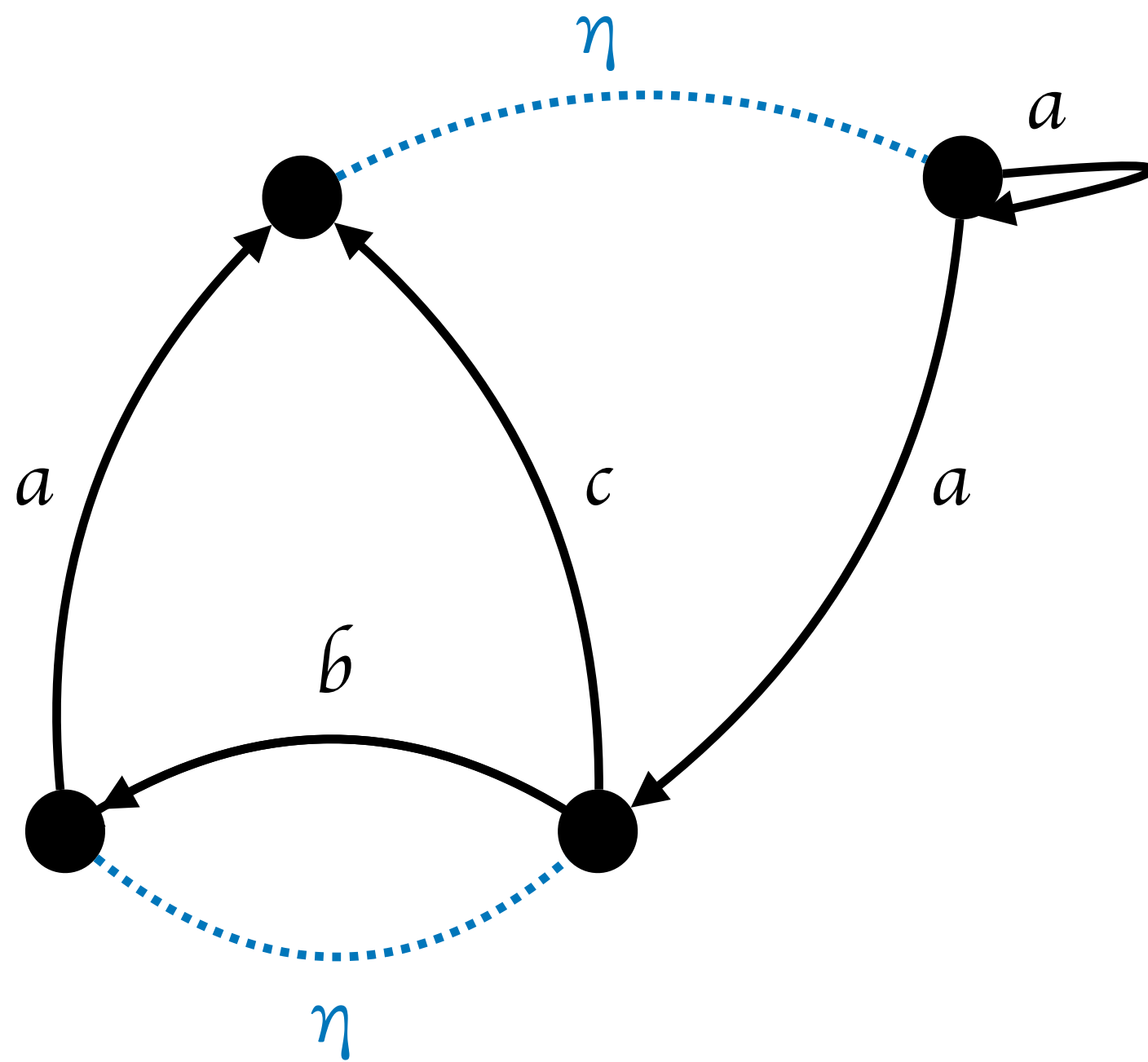
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Corollary Definable transductions are closed under relational composition

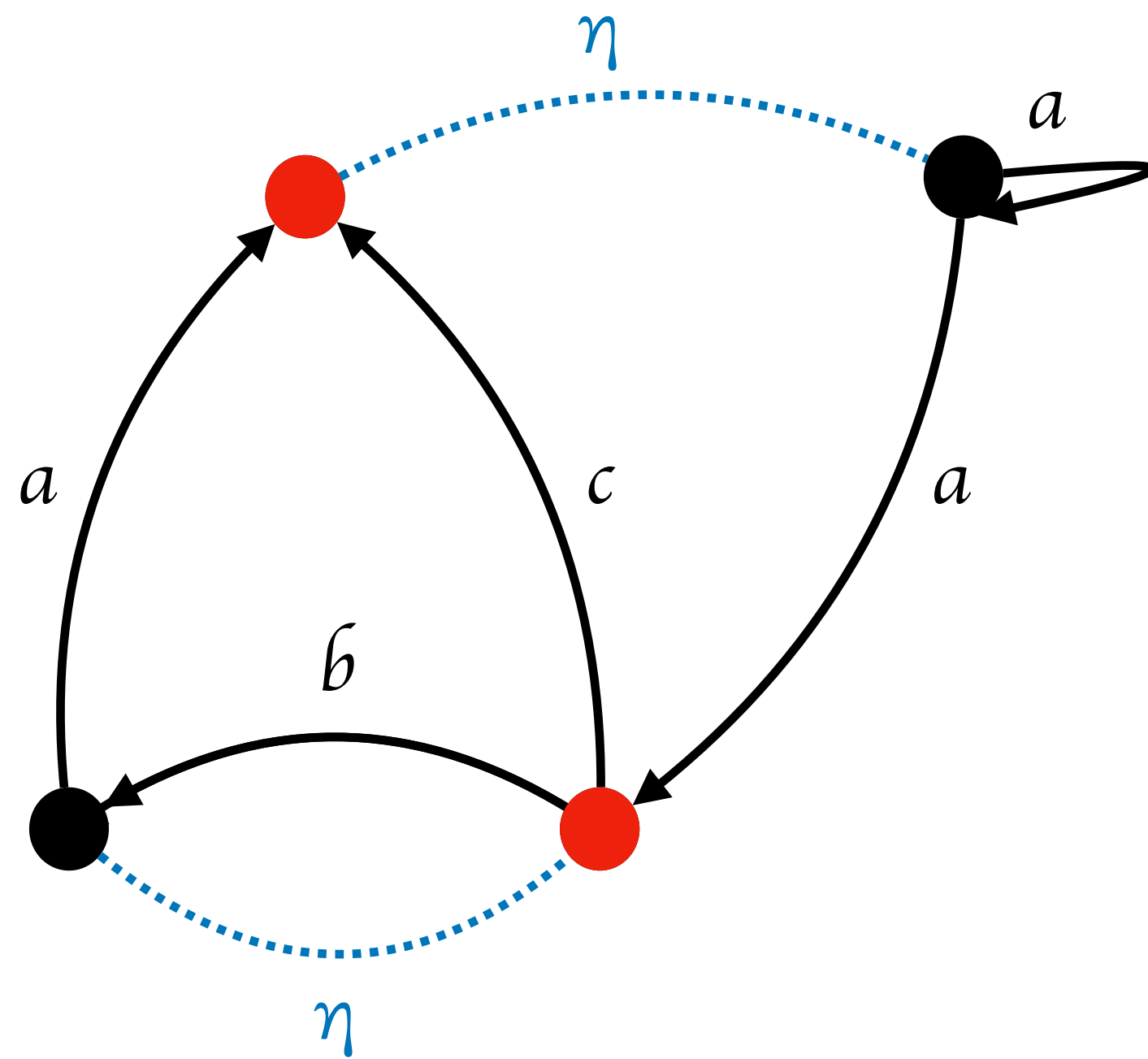
Quotienting is a Definable Transduction

$\eta(x,y)$ = a MSO formula defining an equivalence relation



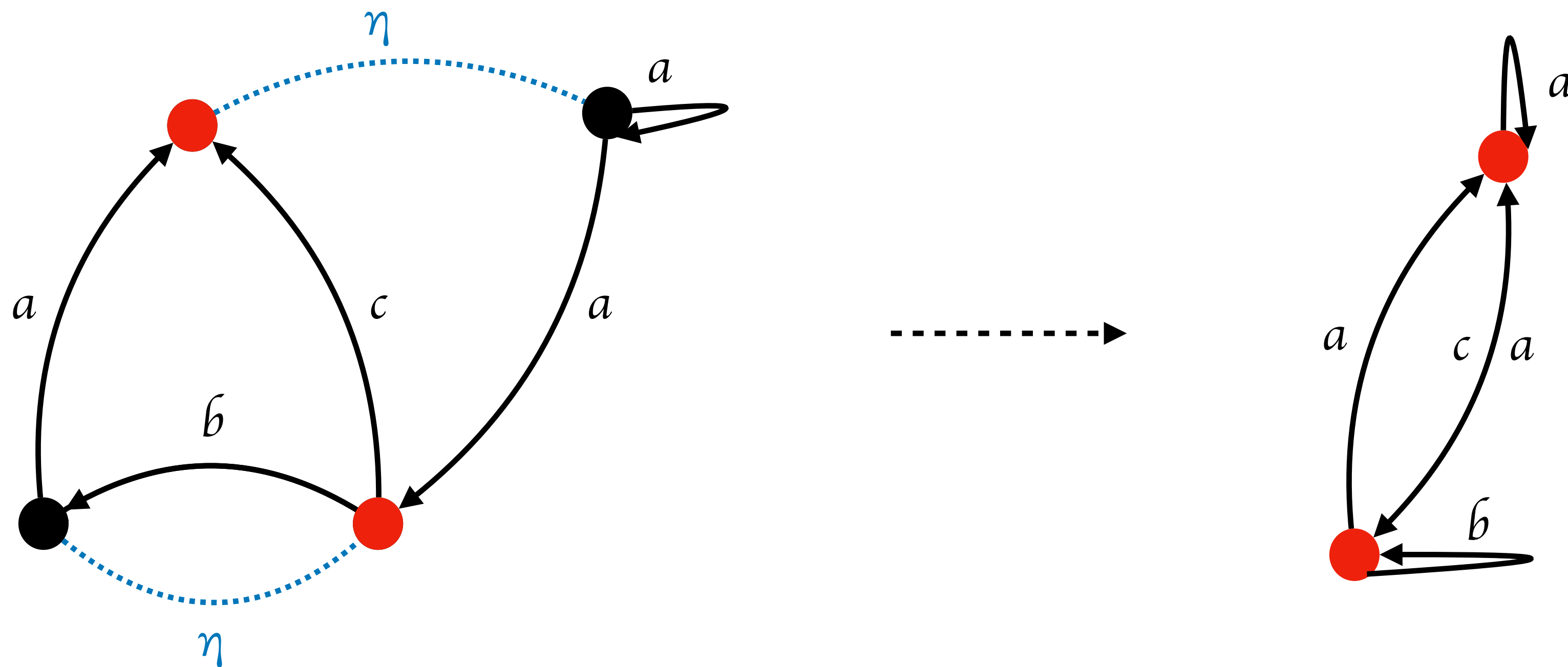
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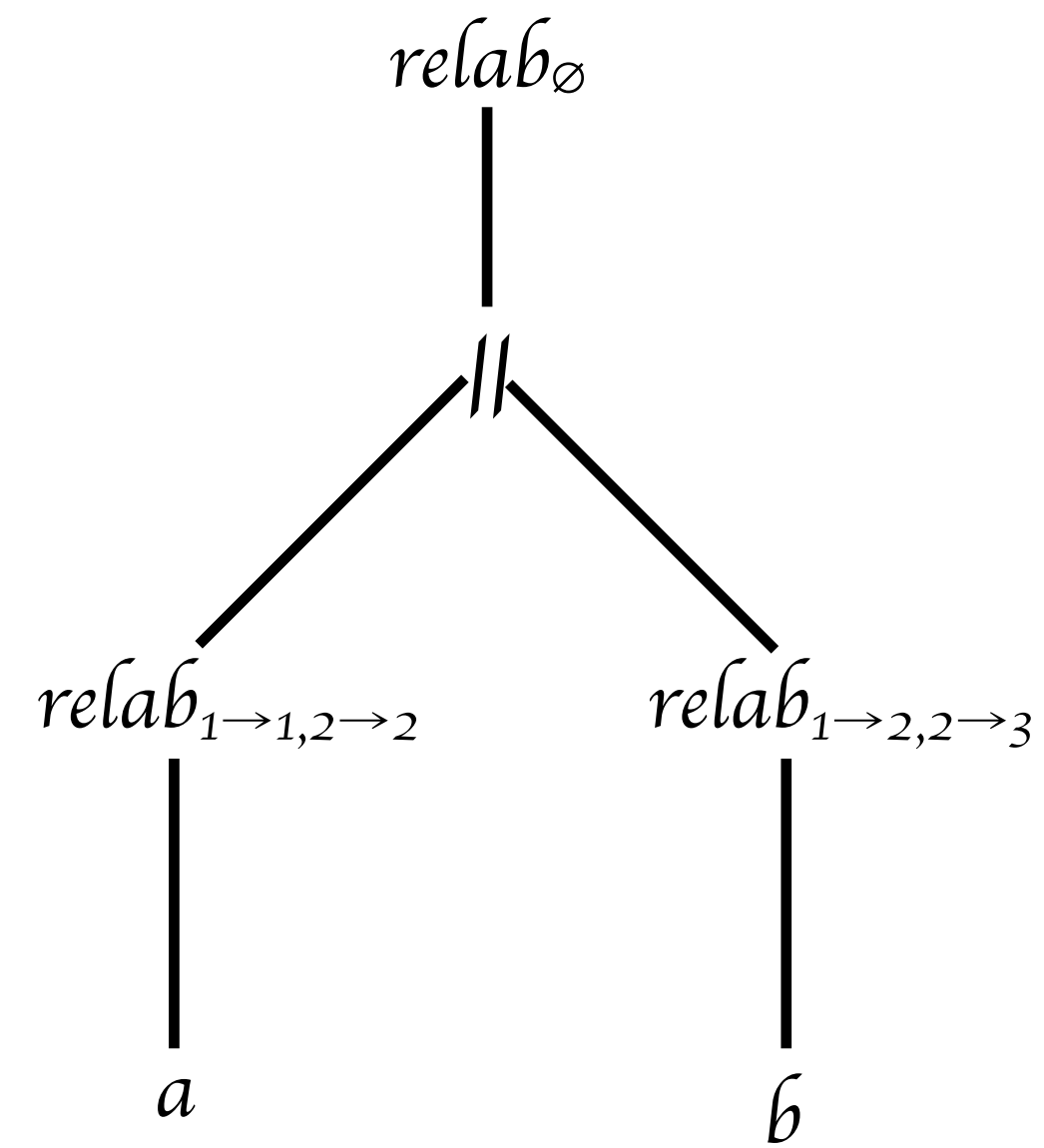


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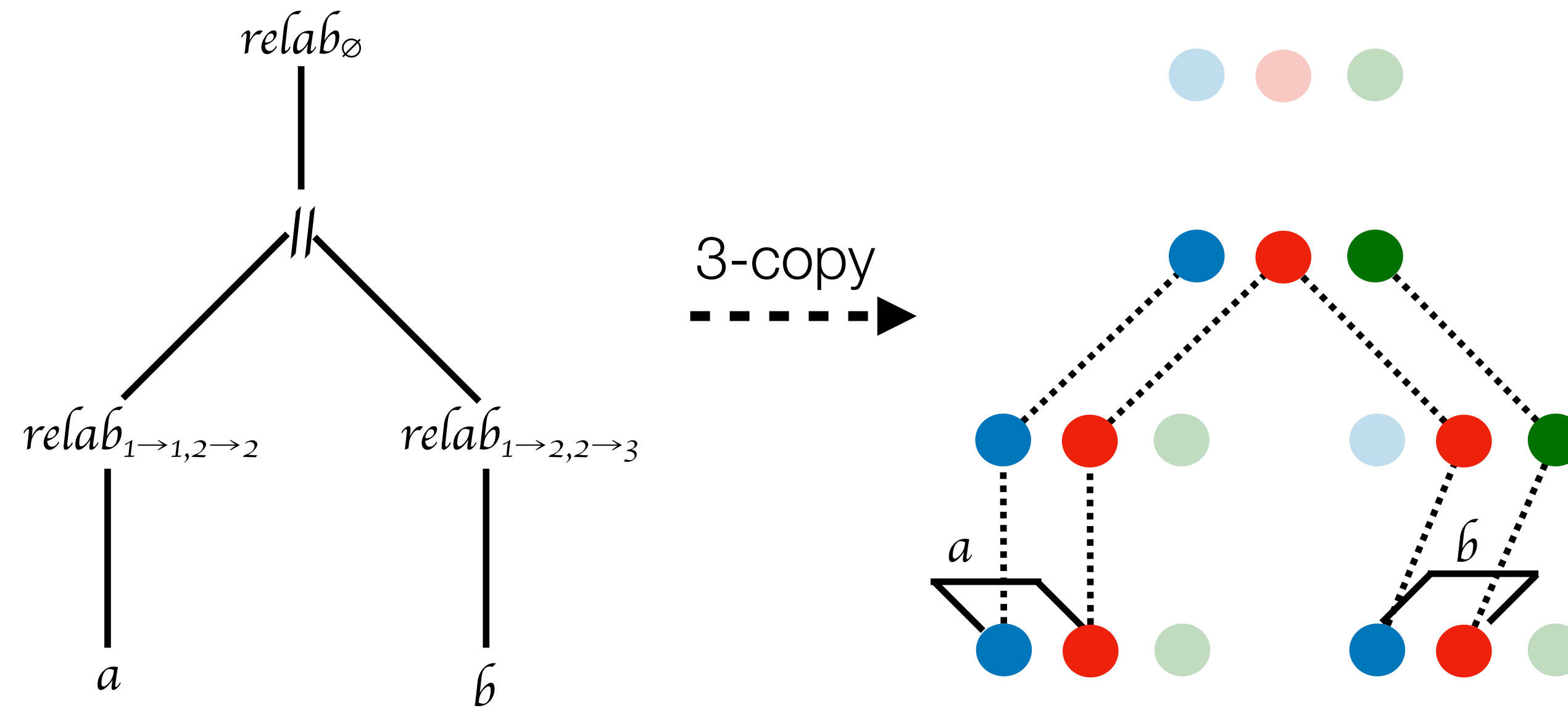
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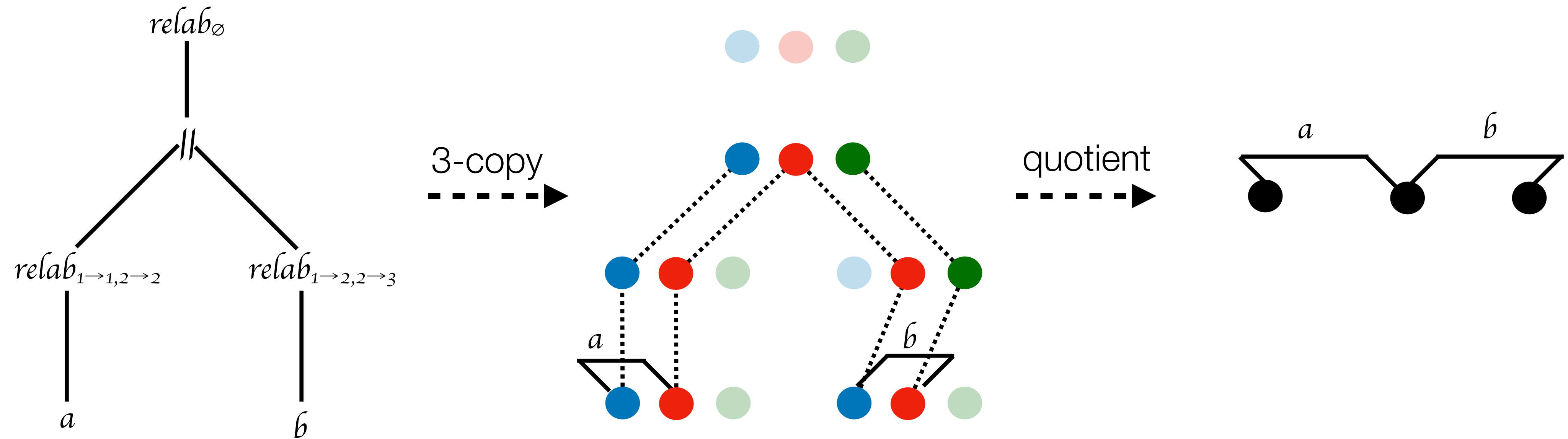
HR-Term Evaluation is a Definable Transduction



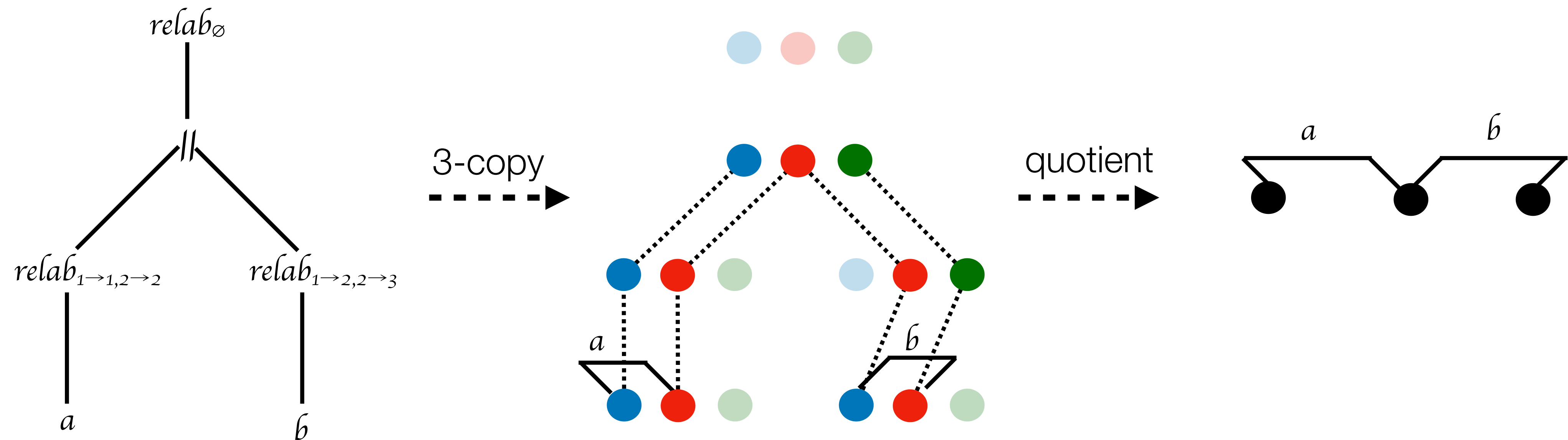
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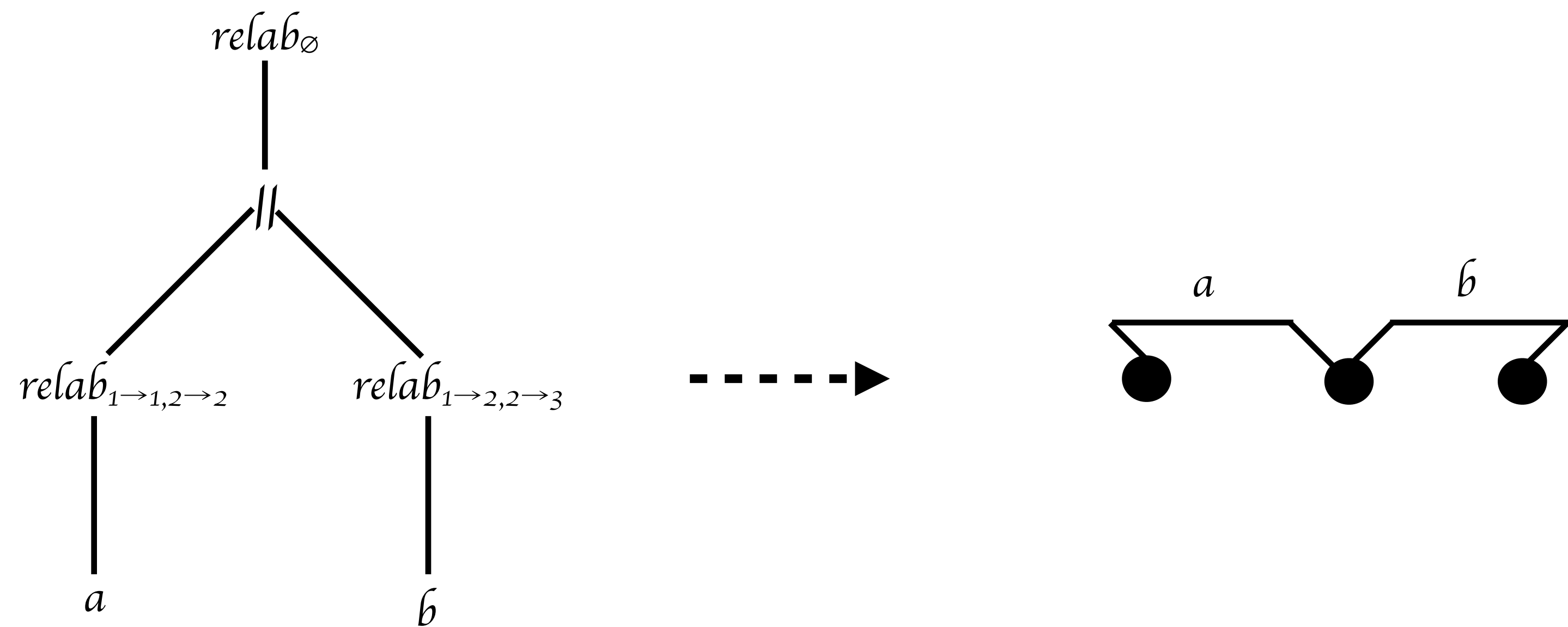
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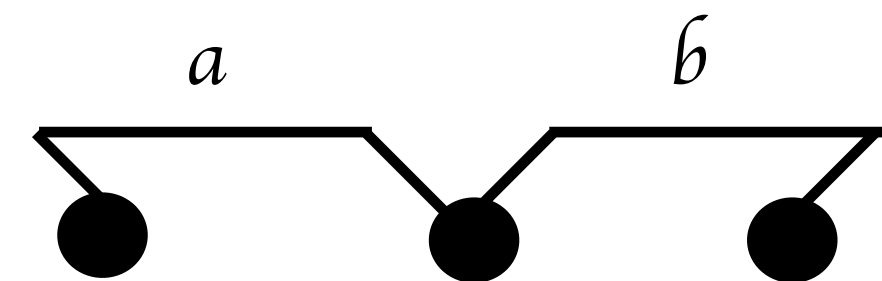
Theorem [Courcelle&Engelfriet] A set $\mathcal{L} \subseteq \mathcal{G}(\Sigma)$ of hypergraphs is HR-context-free $\Leftrightarrow$
 $||\mathcal{L}|| = \delta(||\mathcal{T}||)$, for a recognisable set $\mathcal{T}$ of ranked trees and a definable transduction δ

same for $|\mathcal{T}|$ **(ST)**

Parsable Classes of Hypergraphs

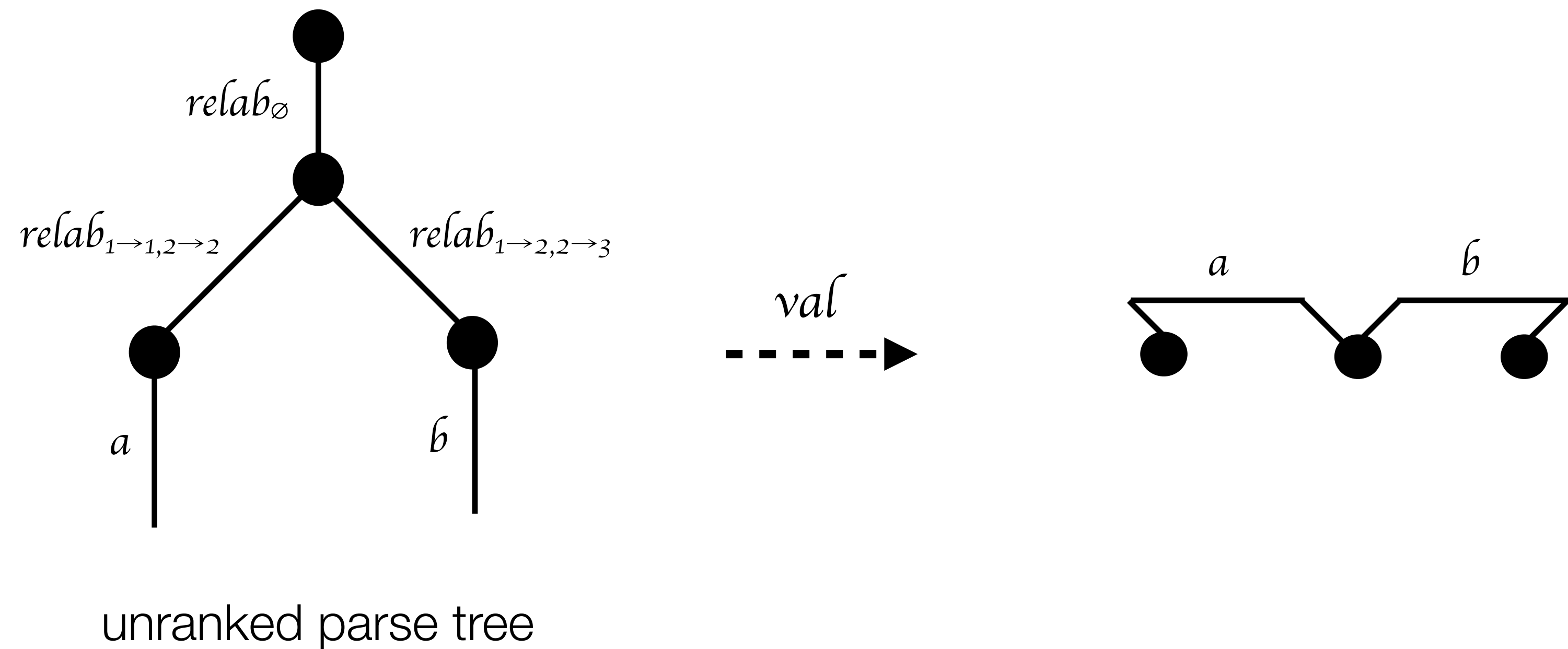


Parsable Classes of Hypergraphs



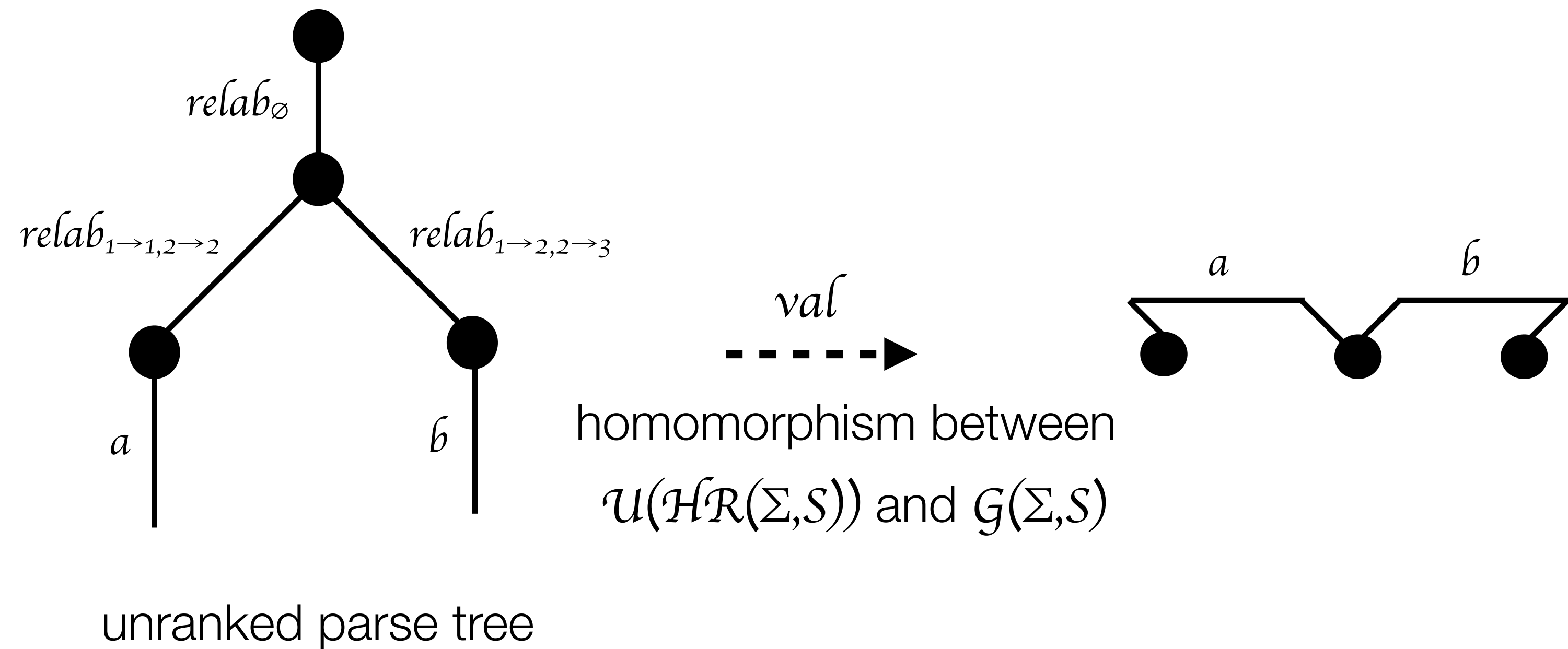
Parsable Classes of Hypergraphs

$\mathcal{HR}(\Sigma, S) \stackrel{\text{def}}{=} \text{finite set of HR operation involving sorts } \tau \subseteq S, \text{ for some finite } S \subseteq_{\text{fin}} \mathbb{N}$



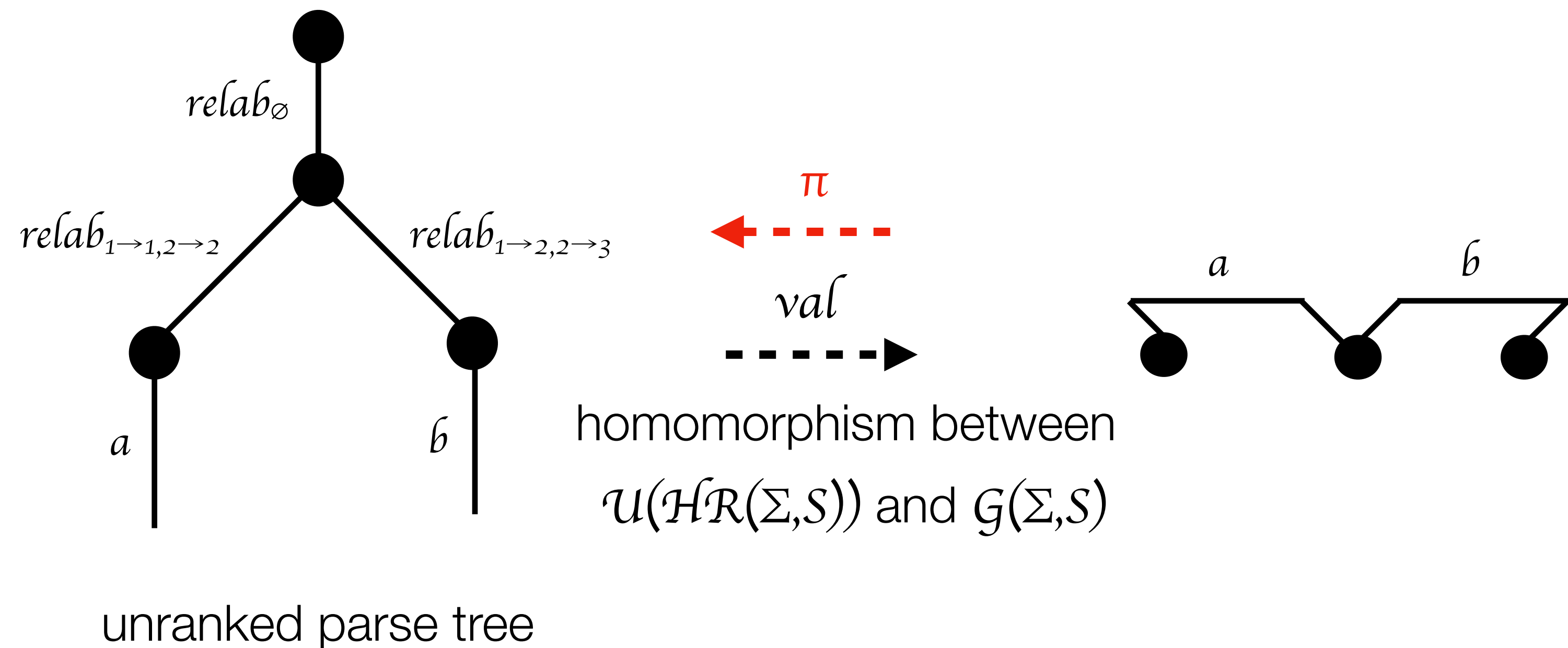
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$C \subseteq \mathcal{G}(\Sigma, S)$ **parsable** $\stackrel{\text{def}}{=} \text{there exists a definable transduction } \pi \subseteq \text{val}^{-1}$
that builds some parse tree for each graph in C

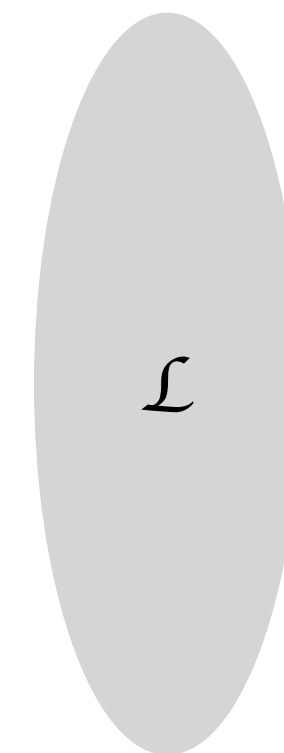
Parsable Classes of Hypergraphs

Theorem [Courcelle] Let C be a parsable class of hypergraphs.

A set $\mathcal{L} \subseteq C$ is recognisable $\Leftrightarrow \mathcal{L}$ is CMSO-definable

" $\Rightarrow$ "

$G(\Sigma, S)$



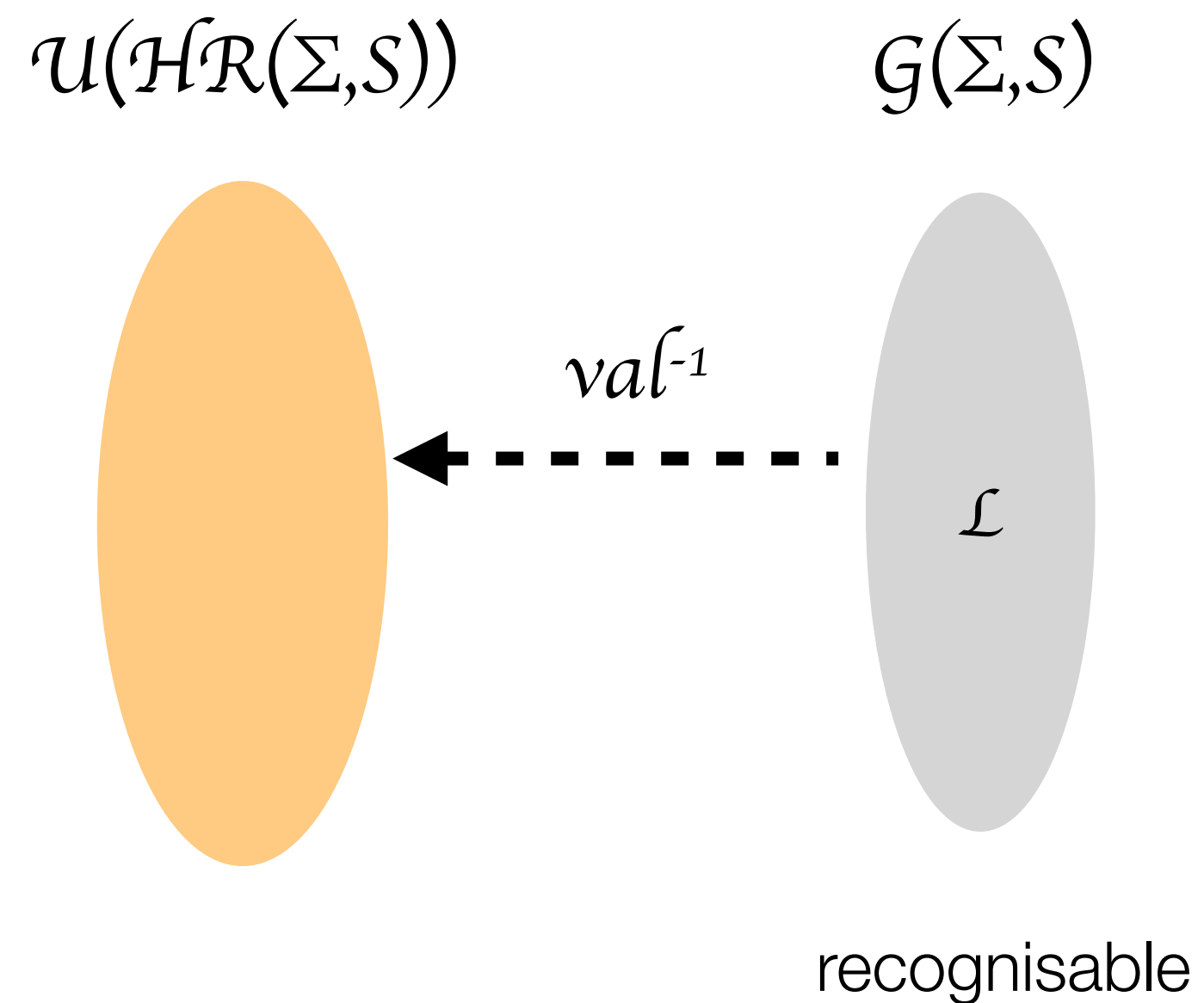
recognisable

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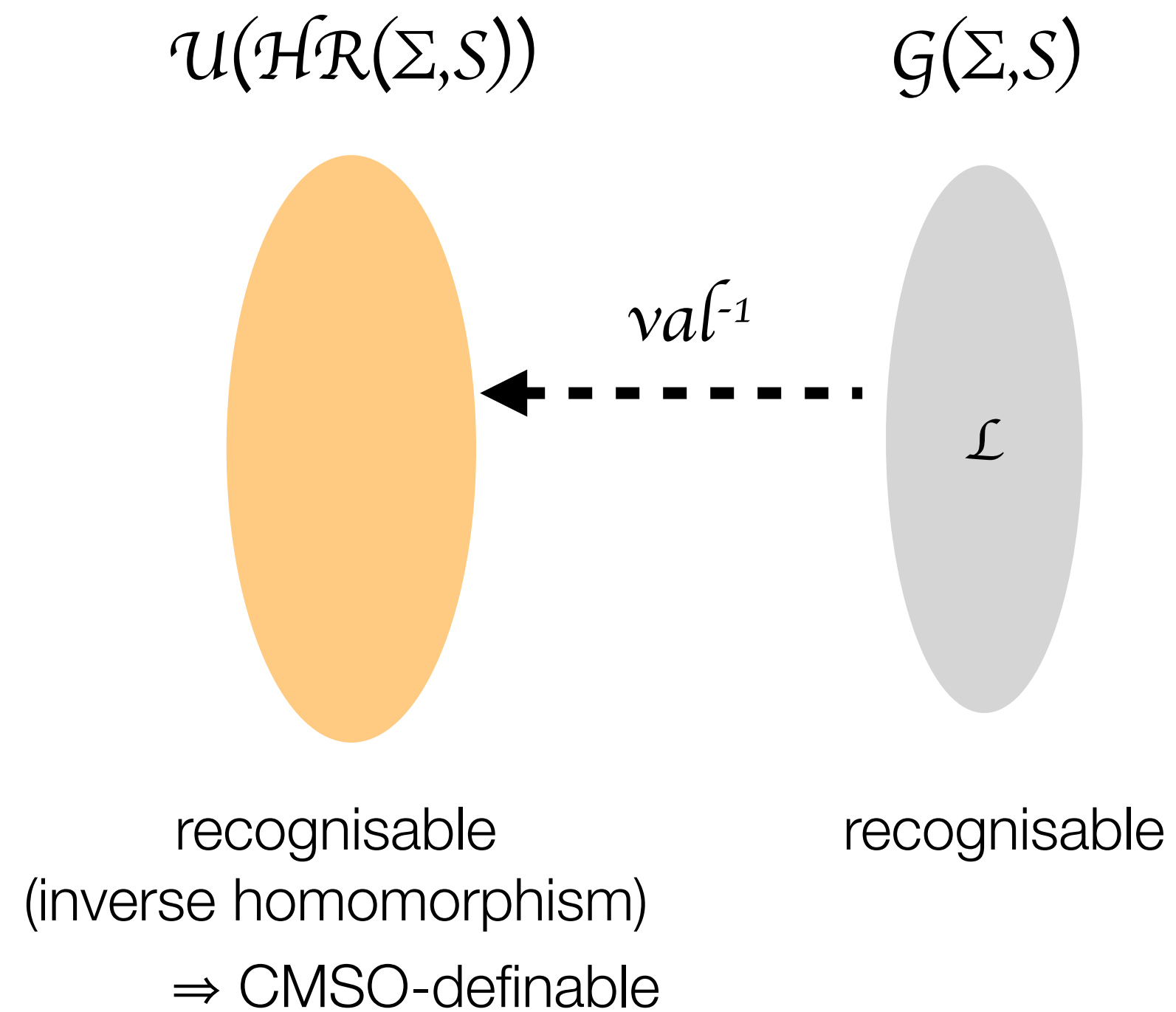


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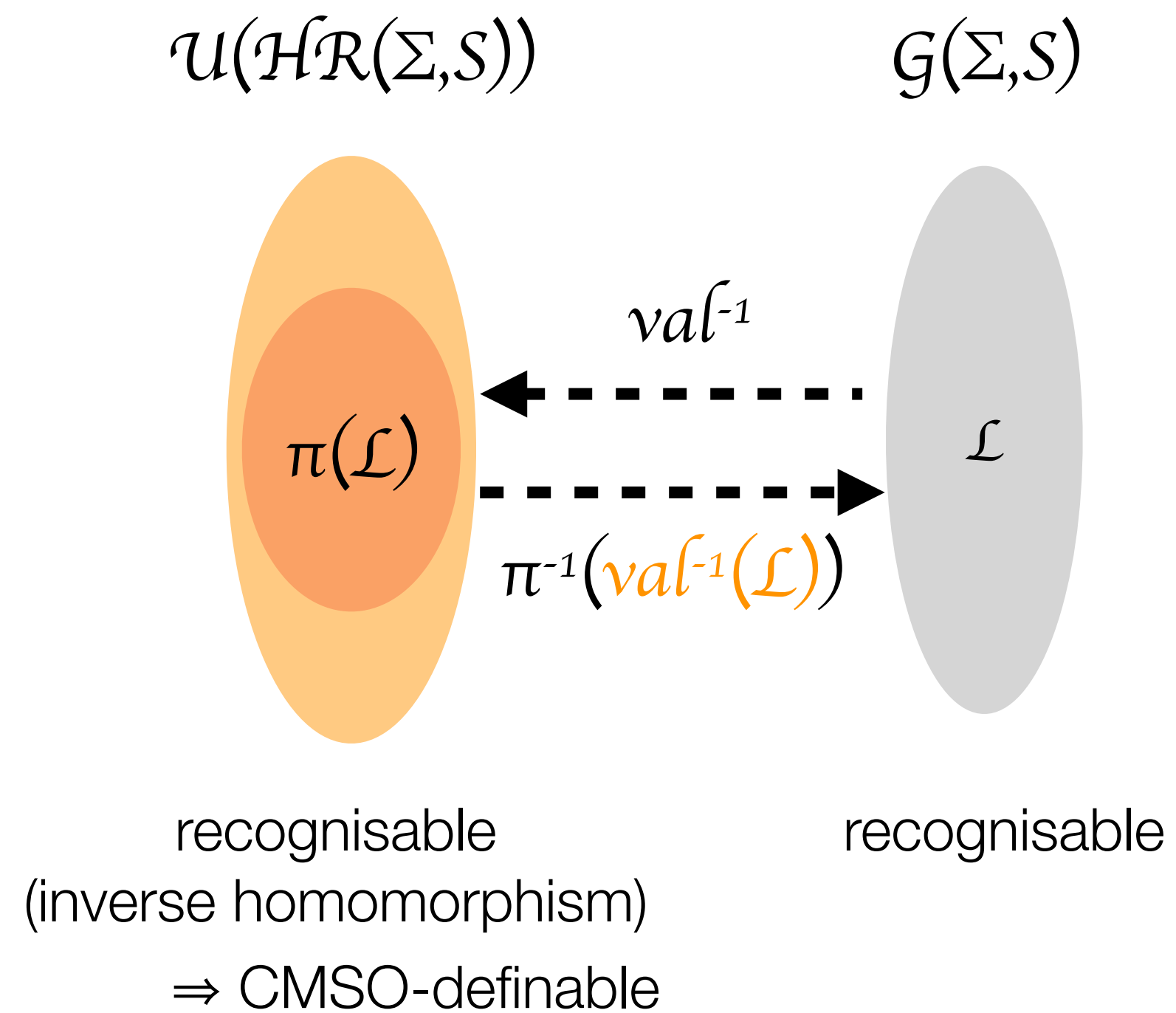


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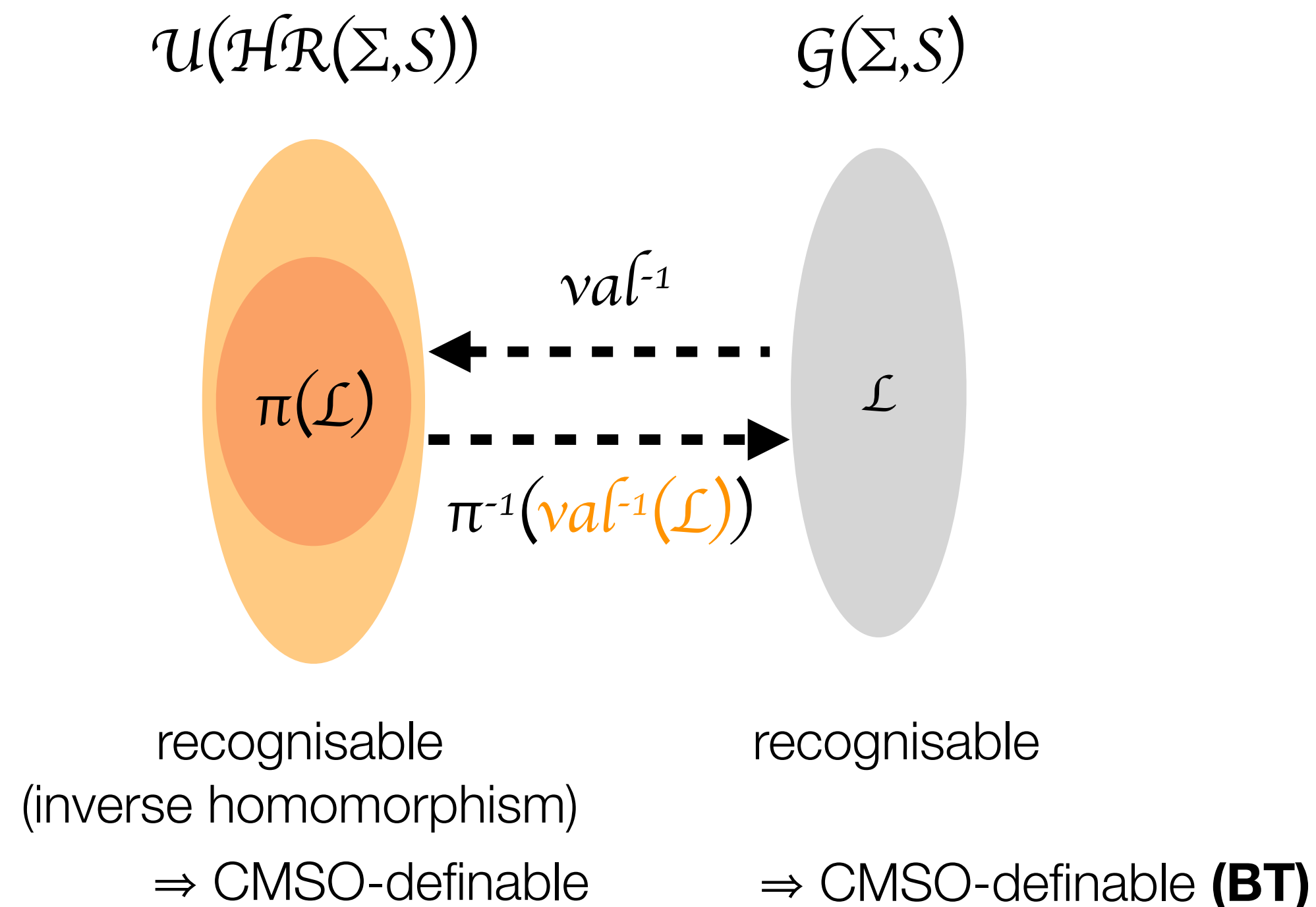


Parsable Classes of Hypergraphs

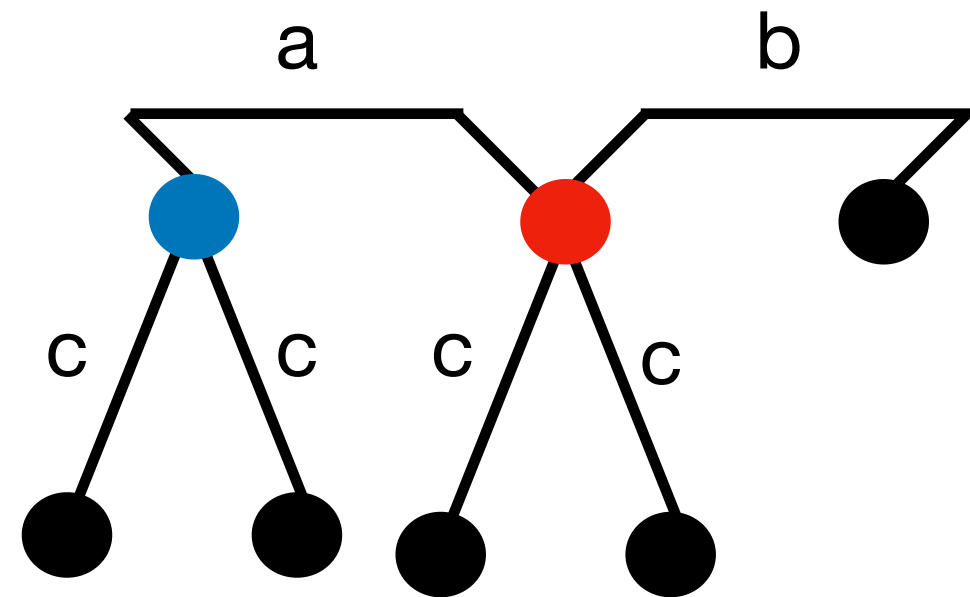
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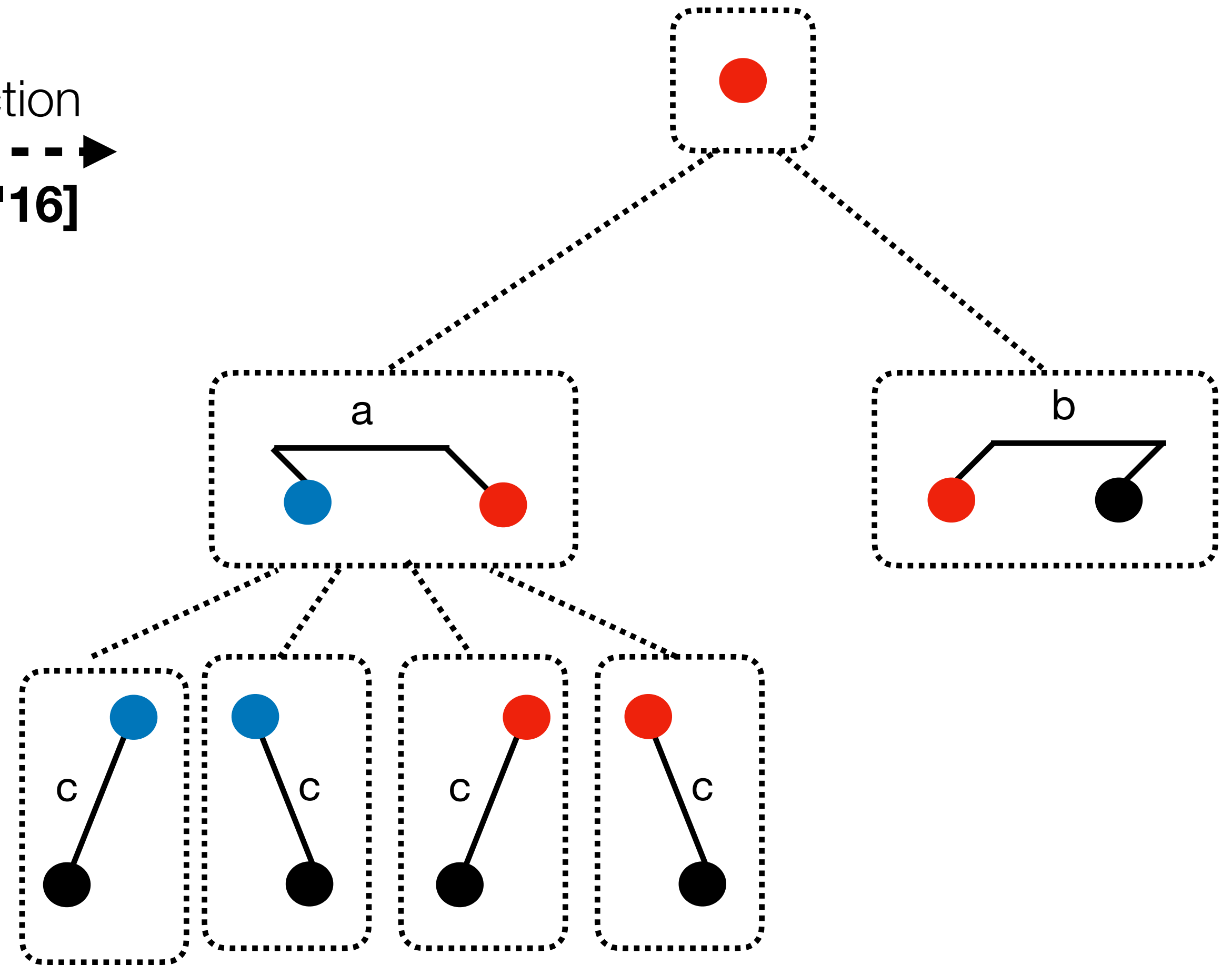
Parsing with Tree Decompositions



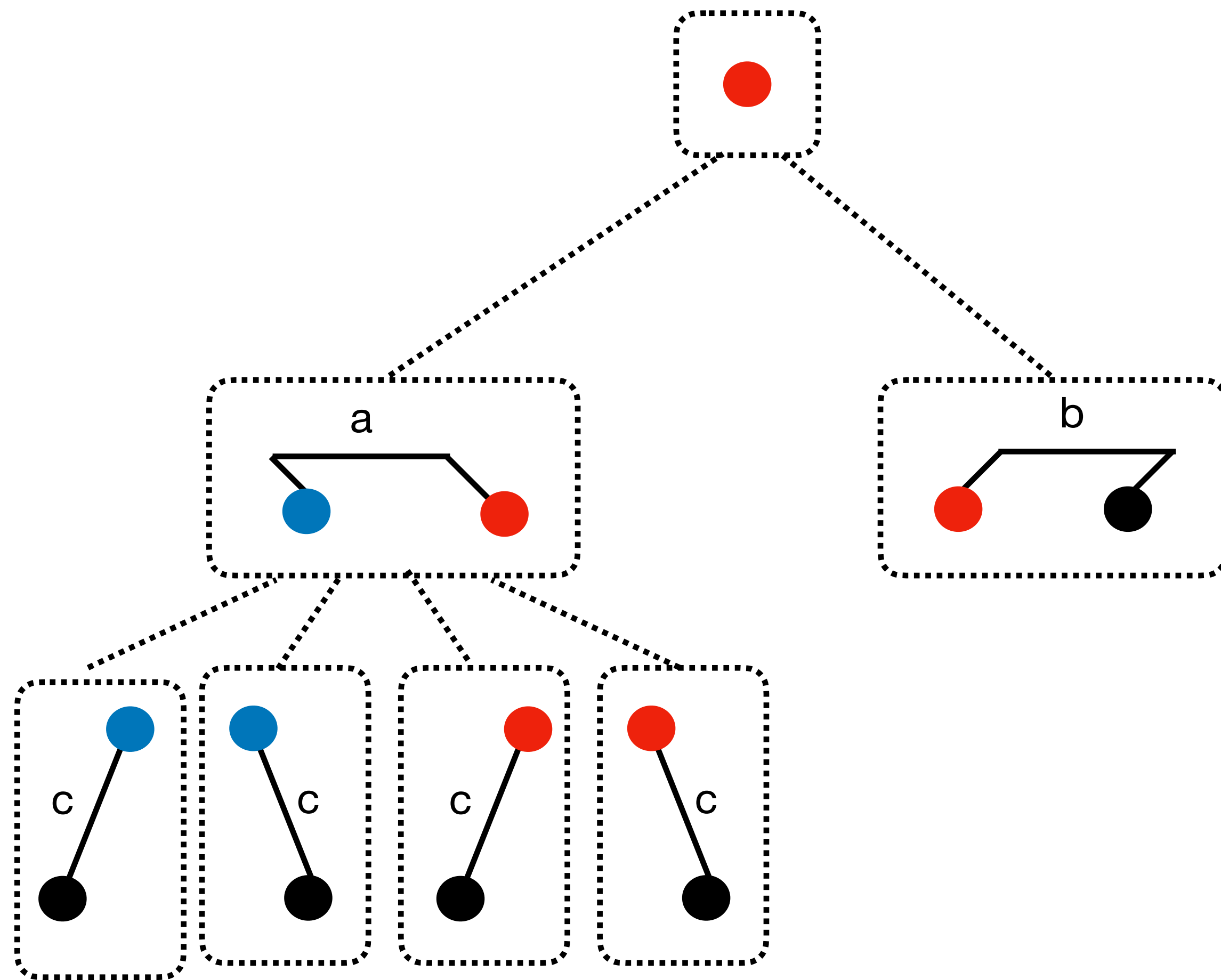
MSO-definable transduction



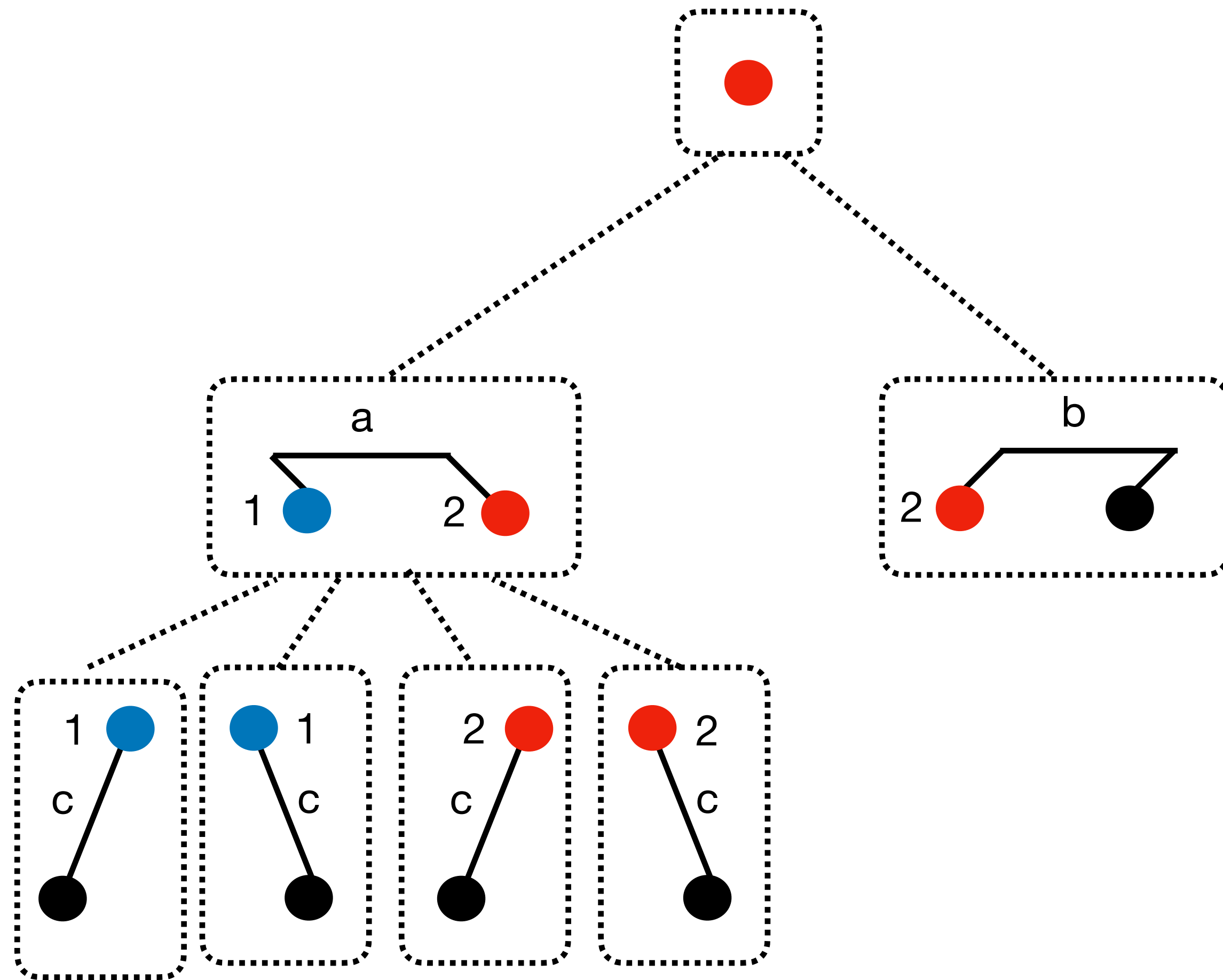
[Bojanczyk&Pilipczuk'16]



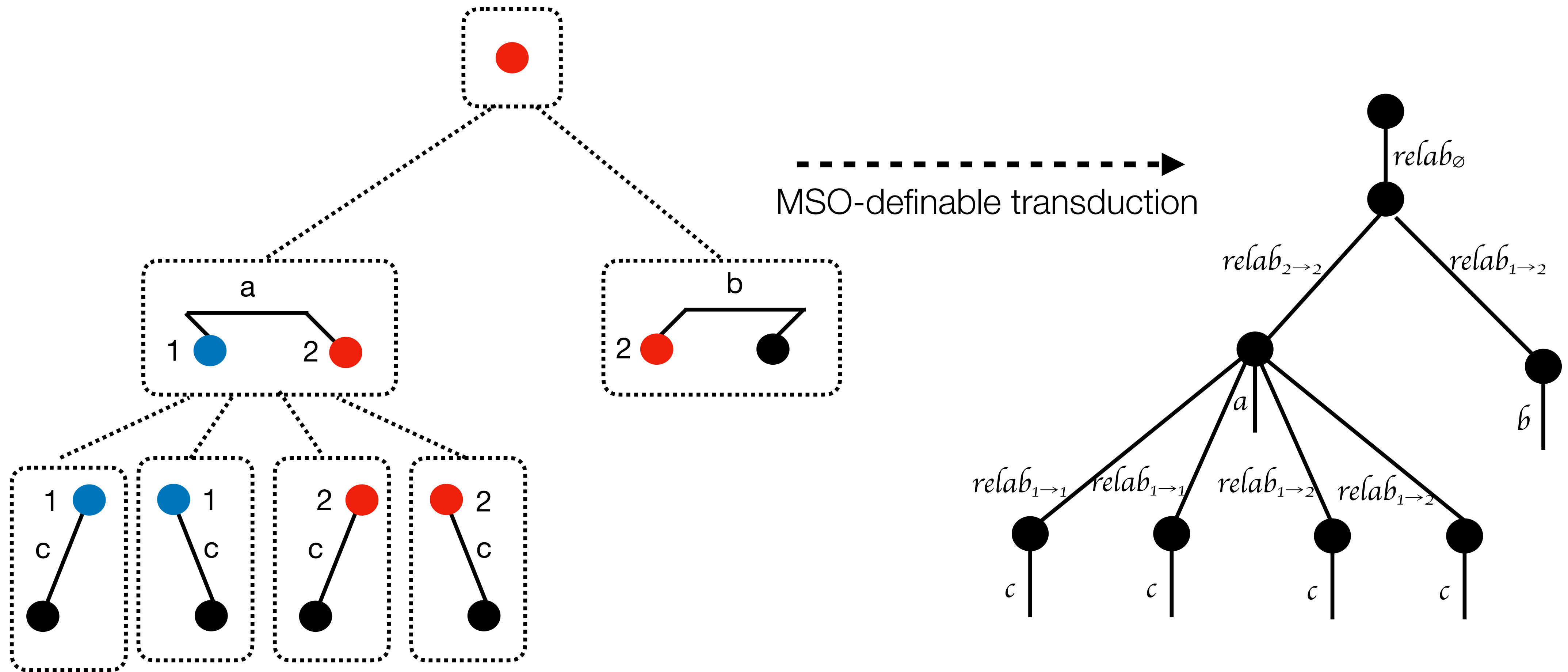
Parsing with Tree Decompositions



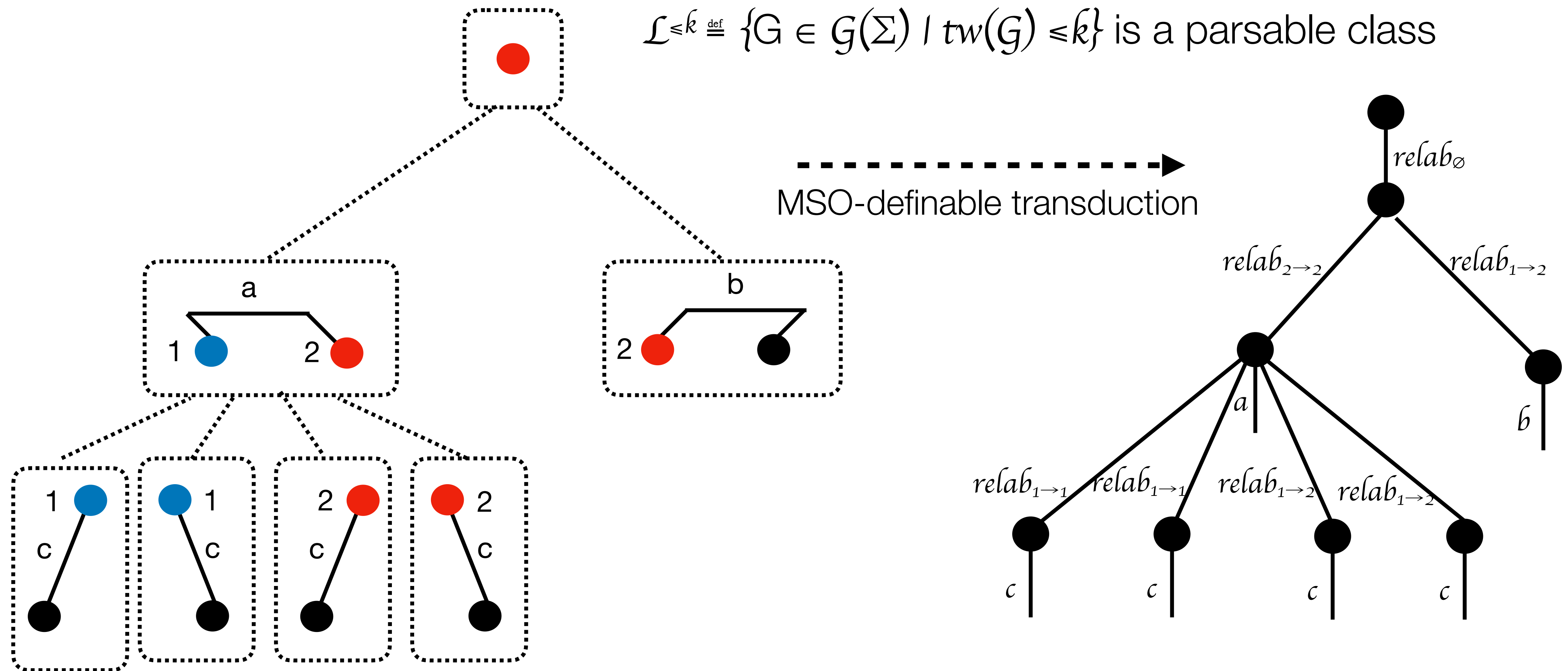
Parsing with Tree Decompositions



Parsing with Tree Decompositions



Parsing with Tree Decompositions



Theorem [Bojanczyk&Pilipczuk'16] A set $\mathcal{L} \subseteq \mathcal{L}^{\leq k}$ is recognisable $\Leftrightarrow \mathcal{L}$ is CMSO-definable